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Department of Creative Technologies

Artificial Intelligence

EZsell

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Bachelor of Science in Artificial Intelligence (2022-2026)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.

Faculty of Computing and AI, Air University, Islamabad
Department of Creative Technologies
Artificial Intelligence

EZsell

A project presented to
AIR University, Islamabad

In partial fulfillment
of the requirement for the degree of

Bachelor of Science in Artificial Intelligence (2022-2026)

By

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DECLARATION

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (AI) "**EZSELL**" was developed by **Muhammad Ahmed (222550), Dawood Haider (221574)** and **Ahmed Abassi (220965)** under the supervision of **Dr. Madiha Yousaf** and that in (their/his/her) opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences / Software Engineering.

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Executive Summary

In the rapidly expanding market for pre-owned products, especially in Pakistan, there is an ongoing task of ensuring fair pricing, verifying product authenticity, and distinguishing between authentic listings and physical authenticity. Traditional markets are often plagued by historical checks, cheat lists, and the "uncertainty factor" where customers can't visualize how things — especially devices — will look from their true surroundings. Relying on book price studies is time-consuming and prone to error, while static images do not provide comprehensive analysis and consistency.

To overcome these challenges and modernize the pleasure of returning, EZSell has evolved. EZSell is an intelligent, AI-powered marketplace designed to automate matching and improve user engagement. It is a complete platform that extracts analytics for individual sellers and shoppers by combining modern machine learning and visualization technologies.
EZSell offers a document-driven approach to antiquities financing through three central platforms:

1. Intelligent Pricing: Using loud voice models (LLM) and real-time market analysis from platforms such as GSMArena and local markets, EZSell provides sellers with statistics-based calculators, avoiding individual overpricing and discounting.
2. Augmented Reality (AR) Integration: Regarding the list of devices, the device allows customers to virtually "try" their own home's attractions. Using WebXR, customers can rotate, rotate, and attach 3-D models to their real floors and components, ensuring the product fits their space before purchasing..
3. Automatic integrity & fraud detection: The tool uses advanced algorithms, such as CLIP (Contrastive Language-Image Pretraining) and dHash (Munhashing), to verify art content and images. This ensures that fraudulent, duplicative, or misleading listings are flagged automatically before they reach the customer.

Technically, EZSell is built as a highly integrated web tool with a FastAPI (Python) back end and a React (Vite) front end. It uses Groq-powered Llama three models for semantic evaluation and Tripo AI for automatic 2D-to-3D model generation. The machine operates on a real-time statistical pipeline that filters the market outlook the use of IQR (Interquartile Range) analysis, ensuring that the insights provided to the user are both accurate and aggressive.

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All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

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Muhammad Ahmed	Ahmed Abassi	Dawood Haider
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Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
AWS	Amazon Web Services
C2C	Consumer-to-Consumer
CLIP	Contrastive Language-Image Pretraining
CORS	Cross-Origin Resource Sharing
CPID	Certificate Profile ID
CPU	Central Processing Unit
CSV	Comma-Separated Values
DB	Database
dHash	Difference Hashing
EC2	Elastic Compute Cloud
FAISS	Facebook AI Similarity Search
GLB / GLTF	Graphics Language Binary / Transmission Format
GPU	Graphics Processing Unit
GSMArena	Global System for Mobile Communications Association Arena
GZip	GNU Zip Compression
HDD	Hard Disk Drive
HTML	HyperText Markup Language
IQR	Interquartile Range
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MDF	Medium-Density Fibreboard
ML	Machine Learning
NLP	Natural Language Processing
ORM	Object-Relational Mapping
OTP	One-Time Password
PBR	Physically Based Rendering
PTA	Pakistan Telecommunication Authority
RAM	Random Access Memory
REST	Representational State Transfer
SMTP	Simple Mail Transfer Protocol
SSD	Solid-State Drive / System Sequence Diagram
SSE	Server-Sent Events
UI	User Interface
UX	User Experience
WebXR	Web Extended Reality

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Abstract

EzSell is a smart, AI-pushed C2C marketplace designed to reverse document asymmetry and transaction friction in Pakistan's traditional electronics and appliance markets. The platform addresses market inefficiencies with a Hybrid Intelligent Pricing Engine that blends real-time smoothed sales records (filtered thru IQR) with special Machine Learning fashions (XGBoost and Gradient Boosting) and LLM-pushed modifications to analyze uncertain variables. To retrieve awesome metadata, an AI-Guided Listing Assistant checks bin descriptions against defined automatic retrieval patterns and forces clients for lack of information. Immersive product visualization is dealt with thru an AR & 3D Visualization Pipeline that converts 2D uploads into 3D fashions using Tripo AI, allowing customers to actually anchor the furnishings of their sows. Finally, the system is secured by a computer vision fraud detection pipeline using CLIP embeddings and perceptual hashing (dHash) to reward refresh or inventory listings, along with a SentenceTransformers-powered Semantic Search gadget, creating a secure, transparent, and equally stable retail store.

1 Introduction

In recent years, the customer-to-customer (C2C) e-commerce industry has witnessed tremendous growth worldwide, pushed by the use of digital retailers and customer behavior movements. In developing countries like Pakistan, this fluctuation is additionally amplified by macroeconomic units. The combination of the recession, the increasing fees for digital innovations, due to austerity, and a developing environmental consciousness has led to a massive upsurge in the pre-owned materials sector. Buying and selling second-hand goods — especially cell phones, computers, and appliances — has evolved from a lust to a financial need for most citizens. As a result, digital media and peer-to-peer markets have become key opportunities for financial exchange.

Despite the overwhelming demand, the current competitive markets in Pakistan, which are dominated by conventional structures including OLX Pakistan and informal platforms such as the Facebook market, suffer from institutional challenges including excessive business competition and data asymmetry. Sellers, lost accountants, struggle to adequately pay for their products, frequently listing them with their arbitrary value. This results in unsold or undervalued inventory. Conversely, retailers face a higher risk by not being able to visually verify the physical quantity or actual placement of products, which is particularly difficult for equipment. The loss of basic pricing models and immersive visual functions severely hampers market performance.

In addition, these older systems suffer from systematic fraud and a continuing lack of accountability. Scammers routinely post replica products, use graphic photos to cover up physical defects, put them up for sale in non-existent equipment, and place them in fake classrooms to avoid privacy. Existing systems often rely on manual reporting or original keywords based on total filters, which can be passed effortlessly. Automatic authentication, computer-vision inspection, and price sensitivity detection are not performed, creating poor compatibility with the environment, mainly due to long communication circuits and security issues. This indicates a strong need for a next-generation technology platform that integrates security automation features without delay in the ad placement pipeline.

To deal with the significant shortcomings, this initiative introduces EzSell, an AI-powered, full-featured C2C marketplace tailored specifically for the Pakistani context. EzSell redefines the peer-to-peer transaction experience by integrating advanced fashion recognition tools, laptop intelligence and prescient leads, natural language processing services, and interactive 3D WebXR visualization engines right straight into a cohesive internet application. At the center of the EzSell system are 5 interconnected smart modules: a Hybrid intelligent pricing engine (the use of XGBoost and Gradient Boosting), a CLIP-first based intelligent fraud protection and validation pipeline, an interactive Tripo AI-powered 3-d AR Vision & Recommendation model, Search engine and recommendations and an intelligent LLaMA trio.1-8b Groq-powered Chatbot assistant.

The primary goal of this final year project is to lay out, develop and validate EzSell's technical architecture, demonstrating how current artificial intelligence technology can be engineered into a production-ready web application to address real-world global advertising problems. The scope of these paintings covers the entire software engineering life cycle, from requirements specification and architectural layout (which include distinctive UML modeling) to gadget deployment to gain knowledge of endpoints and frontend user interfaces. Special emphasis is placed on optimization, which ensures that complex commitments such as 3-d model reconstruction and semantic similarity search run easily inside familiar browser environments.The remainder of this report is organized to present a comprehensive, technical walkthrough of the EzSell project from inception to deployment.

The remainder of this report is organized to present a structured, technical walkthrough of the EzSell project from inception to deployment. Chapter 1: Introduction establishes the vision, provides a comparative related system analysis, outlines project deliverables, lists system limitations, and reviews the technology stack and relevance to the academic curriculum. Chapter 2: Problem Definition defines the precise inefficiencies of the second-hand C2C Pakistani marketplace, outlines the proposed dual-track AI solution, sets system objectives, maps the scope, and introduces the eleven core system modules. Chapter 3: Requirement Analysis details user classes, outlines requirement identifying techniques (including use case diagrams, detailed use case tables, and event-response tables), defines functional and non-functional requirements (reliability, usability, performance, and security), and specifies external user, software, hardware, and communication interfaces. Chapter 4: Design and Architecture presents the layered system architecture, details UML design models (activity diagrams, class diagrams, system sequence diagrams, and state transition diagrams), defines the database schema through a data dictionary, and showcases the human-interface mockup screens. Chapter 5: Implementation explains the core algorithms for the eleven modules, reviews external API and SDK integrations, and documents the cloud deployment environment (compute, storage, network, and access settings). Chapter 6: Testing and Evaluation presents comprehensive API endpoint tests, unit tests, functional validations, and integration tests to verify the system. Chapter 7: Conclusion and Future Work summarizes the contributions of EzSell and outlines roadmap items. Finally, Chapter 8: References lists the academic and technical bibliography cited throughout this work.

1.1 Vision Statement

For individual buyers and sellers in Pakistan who face challenges in pricing second-hand electronics and furniture accurately and safely, EzSell is an AI-powered web marketplace that provides real-time price intelligence, automated fraud prevention, AR visualization, and a personalized buying experience. Unlike existing platforms such as OLX and Facebook Marketplace, which lack AI-assisted pricing, structured fraud detection, and immersive product visualization, EzSell leverages a multi-model ML ensemble, a Groq LLM layer, CLIP-based image validation, SentenceTransformer semantic recommendations, and Tripo AI 3D generation to deliver a trustworthy, data-driven, and visually superior marketplace for the pre-owned goods economy of Pakistan.

1.2 Related System Analysis / Literature Review

The following table presents a comparative analysis of leading marketplace platforms against the proposed EzSell solution, identifying their weaknesses and the specific innovations EzSell introduces.

Table 1 Related System Analysis with Proposed Project Solution

Application Name	Weakness	Proposed Project Solution
OLX Pakistan	No AI-assisted pricing; prices are seller-determined leading to widespread overpricing. High incidence of scam listings with no automated image verification.	EzSell integrates a dual-track ML + LLM pricing pipeline (XGBoost, LightGBM, Groq) with live OLX market scraping. CLIP-based image validation eliminates fraudulent listings.
Facebook Marketplace	Lacks structured pricing guidance. No category-specific fields for electronics. Buyers cannot visualize furniture in their actual space.	EzSell provides category-specific attribute fields, AI price suggestions, and a full AR Try-On viewer powered by Tripo AI and Google model-viewer for in-room furniture visualization.
Daraz.pk	Primarily a new-goods B2C platform; does not support C2C used-goods listings. No secondhand ecosystem.	EzSell is purpose-built for the secondhand C2C market, supporting individual sellers with full listing lifecycle management and AI-assisted used-goods pricing.

1.3 Project Deliverables

The following deliverables constitute the complete EzSell 1.0.0 production release:

- D-1: Fully functional FastAPI backend with RESTful APIs for all marketplace operations.
- D-2: React 18 + Vite + TypeScript frontend SPA with Tailwind CSS and Shadcn UI.
- D-3: AI-powered price prediction engine combining a trained ML ensemble and Groq LLM validation.
- D-4: Automated fraud prevention pipeline utilizing CLIP category validation, dHash duplicate detection, and scam keyword filtering.
- D-5: Semantic recommendation engine using SentenceTransformer embedding-based personalized feeds.
- D-6: AR/3D furniture try-on module featuring Tripo AI GLB generation and WebAR viewer integration.
- D-7: Admin analytical dashboard equipped with listing moderation, user management, and platform engagement metrics.
- D-8: Real-time in-app messaging system providing direct buyer-to-seller communication.
- D-9: AI chatbot assistant (EzSell Assistant) powered by Groq llama-3.1-8b-instant with live context injection.
- D-10: Cloudinary-integrated cloud media storage with optimized content delivery.

1.4 System Limitations / Constraints

- L-1: EzSell is a web-only platform at launch; no dedicated iOS or Android native application is provided, though the frontend is fully mobile-responsive.
- L-2: Only three product categories are supported in v1.0.0 — Mobiles, Laptops, and Furniture.
- L-3: Web scraping of OLX Pakistan via DuckDuckGo is subject to external rate limits; the system falls back to cached CSV data if rate-limited.
- L-4: The platform does not facilitate in-app payment processing; all financial transactions are completed offline between the buyer and seller.
- L-5: AR 3D generation via the Tripo AI API requires significant compute time, taking approximately 30–90 seconds per generation depending on third-party API availability.

1.5 Tools and Technologies

Table 2 below lists all hardware and software tools, grouped by category, used in the implementation of EzSell.

Table 2 Tools and Technologies for EzSell

Tool / Technology	Version	Rationale
Programming Language		
Python	3.11+	Backend logic, ML pipeline, AI services
TypeScript	5.8	Type-safe frontend development
JavaScript	ES2023	Frontend runtime interactions
Frontend		
React	18.3	Component-based SPA architecture
Vite	5.4	Ultra-fast HMR dev server and bundler
Shadcn UI	Latest	Accessible headless UI primitives
Tailwind CSS	3.4	Utility-first styling
3D / AR		
Google model-viewer	4.1	WebAR GLB rendering
Three.js / React-Three-Fiber	0.182 / 8.18	Procedural 3D scenes and WebGL
Backend		
FastAPI	0.115+	Async Python REST API
Uvicorn	Latest	Production ASGI server
SQLAlchemy / Pydantic	2.x	ORM and schema validation

Database		
PostgreSQL / SQLite	15 / 3.x	Production and Dev data storage
AI / ML		
XGBoost / LightGBM	2.x	Primary gradient boosted price prediction (35% weight each)
Scikit-learn	1.x	Random Forest (15%) and GradientBoosting (15%) price prediction
SentenceTransformers	3.x	Semantic embedding model
Groq API	Latest	llama-3.3-70b (pricing); llama-3.1-8b (chatbot)
OpenAI CLIP	Latest	Zero-shot image classification
Tripo AI API	V2	Image-to-3D GLB generation
Infrastructure		
AWS EC2	c7i-flex.large	Cloud compute server for backend API
Cloudinary	Latest	Cloud media CDN and transformation

1.6 Relevance to Course Modules

EzSell integrates extensive knowledge from the core and elective modules of the BS Artificial Intelligence (Fall 2022) curriculum at Air University. The system directly applies theoretical concepts into a production-ready application:

- AI332 Machine Learning: Directly applied in the dual-track pricing engine, utilizing ensemble methods (XGBoost, LightGBM, Random Forest, Gradient Boosting) to train predictive models on scraped second-hand market data.
- AI320 Natural Language Processing: Implemented for semantic analysis within the multi-layer fraud pipeline (scam keyword detection) and the Groq LLM-powered conversational chatbot assistant.
- AI303 Computer Vision: Utilized for the zero-shot image classification pipeline (OpenAI CLIP) to validate product categories and dHash perceptual hashing to identify duplicate or stolen listing photos.
- SE100 Software Engineering & CS112 Object Oriented Programming: Formed the foundation of the project’s SDLC, architectural design, and the object-oriented structure of the FastAPI backend and React frontend SPA.

- CS230 Database Systems: Applied in designing the normalized PostgreSQL relational database schema and mapping complex polymorphic relationships using the SQLAlchemy ORM.
- CS415 Information Security: Implemented to secure user data via bcrypt password hashing, JWT-based stateless session management, and robust API endpoint protection.
- CS214 Data Structures and Algorithms: Essential for developing the hybrid semantic recommendation engine, optimizing cosine similarity mathematical operations, and executing rapid in-memory vector lookups.
- AI301 Knowledge Representation and Reasoning: Applied in the logic governing the rule-based decision matrix of the fraud detection pipeline and the semantic context injection for the LLM.
- AI443 Virtual and Augmented Reality (AI Elective): Directly applied in the implementation of the 3D furniture try-on module, converting 2D images to GLB models via Tripo AI and rendering them in physical space using WebAR.

2 Problem Definition

Defining the boundaries, constraints, and parameters of a technical problem is an essential precursor to engineering a robust software solution, particularly one that seeks to address complex socio-economic interactions. This chapter provides a comprehensive and academically rigorous definition of the precise problem space that EzSell targets within Pakistan's peer-to-peer (C2C) pre-owned goods market. Peer-to-peer commerce involving used mobile phones, laptops, and furniture represents a large yet highly fragmented sector of the informal economy. Transactions within this sector have historically been characterized by high transaction friction, cognitive fatigue, safety anxieties, and severe information asymmetry. By systematically defining these problems, we establish technical and business justifications for developing an AI-powered marketplace.
To establish a clear structural framework, this chapter is divided into four key sections. First, the **Problem Statement** contextualizes the market inefficiencies, detailing how the lack of automated validation and pricing intelligence leads to a chronic buyer-seller trust deficit, pricing confusion, and financial losses. Second, the **Problem Solution** presents the proposed software solution, articulating how EzSell's integrated machine learning models, computer vision filters, natural language processing keyword scans, and WebXR augmented reality pipelines operate in unison to restore transaction security and confidence. Third, we establish a set of **Measurable Business and System Objectives** that define the operational benchmarks the system must achieve. These objectives include quantitative metrics such as reaching over 90% accuracy in used laptop price predictions, maintaining automated fraud detection false-positive rates below 2%, and achieving page load speeds under 1.5 seconds. Finally, the **System Scope** delineates the functional and non-functional boundaries of the platform. This defines what is supported—such as user-guided listing validations, interactive WebXR simulations for furniture, and in-app secure chat—and establishes the boundaries of what is excluded, such as payment gateways, third-party delivery shipping integrations, and general inventory categories outside the core focus areas.

2.1 Problem Statement

2.2 Problem Statement

The Pakistani second-hand goods market suffers from systemic inefficiencies due to a lack of automated pricing guidance, structural trust mechanisms, and spatial visualization tools, which collectively fuel a chronic buyer-seller trust deficit, high cognitive load, rampant fraudulent listings, and costly physical mismatch anxiety.

2.3 Problem Solution

Many buyers and sellers in the Pakistani second-hand market have requested a structured digital ecosystem that eliminates the guesswork, anxiety, and suspicion inherent in traditional classified advertisement platforms. To address the chronic issue of pricing intuition that leads to financial loss or prolonged sales, the EzSell system implements a dual-track artificial intelligence pricing engine. This engine combines a localized Machine Learning ensemble with real-time Groq LLM market scraping to automatically suggest fair market values, thereby saving sellers time and drastically reducing their cognitive load. By providing these objective pricing benchmarks, the system also protects buyers from inflated prices, subsequently lowering their decision fatigue and purchasing anxiety. Furthermore, to combat the rampant issue of fraudulent listings that exploit the current lack of structural trust, the application deploys an automated, multi-layer fraud prevention pipeline. By seamlessly utilizing zero-shot CLIP image classification and dHash perceptual deduplication upon image upload, the system proactively intercepts stolen photos and category mismatches before they ever reach the public feed. The integration of Natural Language Processing to scan for scam-baiting keywords ensures that the marketplace remains a secure environment, ultimately restoring buyer confidence. For the specific pain points of purchasing second-hand furniture, the future ability for buyers to visualize items in their actual living spaces is achieved through an Augmented Reality try-on module. Utilizing Tripo AI for 3D GLB generation, this feature completely removes spatial anxiety, saving buyers from costly mismatches in size and aesthetic style. Finally, by incorporating a SentenceTransformer semantic recommendation engine, the platform eliminates search fatigue, ensuring that users are presented with highly relevant listings tailored to their behavioral patterns. Collectively, these integrated technologies transform a high-friction, low-trust environment into an intelligent, secure, and psychologically reassuring digital marketplace.

2.4 Objectives of the Proposed System

- BO-1: Deliver AI-assisted price prediction utilizing an ML ensemble and LLM scraping, achieving 85%+ accuracy within ±20% of actual market prices.
- BO-2: Reduce fraudulent listing submissions by at least 70% through a five-stage automated pipeline including image classification and deduplication.
- BO-3: Enable immersive, browser-native WebAR furniture visualization without requiring the user to download a dedicated mobile application.
- BO-4: Streamline the product discovery process by serving personalized, semantic-based feed recommendations in under 800 milliseconds per query.
- BO-5: Facilitate secure and instant buyer-to-seller communication through a real-time, in-app messaging architecture.

2.5 Scope

The scope of the EzSell project encompasses the design, development, and deployment of a web-based, AI-augmented Customer-to-Customer (C2C) marketplace specifically optimized for the Pakistani economy. The boundaries of the platform are strictly defined to facilitate the trading of three distinct product categories: used mobile phones, laptops, and furniture. Within this domain, the system manages the complete digital lifecycle of a product listing, beginning with a streamlined, category-specific seller onboarding process. As part of its core functionality, the system integrates a dual-track AI pricing engine that utilizes a local machine learning ensemble alongside real-time Large Language Model (LLM) web scraping to generate objective market price estimates. To ensure marketplace integrity, the scope includes an automated, multi-layer fraud prevention pipeline that actively scans all user uploads. This pipeline utilizes zero-shot computer vision classification to verify category alignment, perceptual hashing to block duplicate images, and natural language processing to flag scam-baiting terminology in descriptions. For prospective buyers, the platform's range extends to a personalized discovery feed powered by semantic embedding recommendations that match user behavior to relevant listings. A major functionality within the system's boundary is the Augmented Reality (AR) try-on module, which converts 2D furniture images into 3D models via a third-party API and renders them directly in the browser's spatial viewer. To facilitate negotiations, the scope includes a real-time, in-app messaging architecture and an intelligent chatbot assistant configured to provide contextual marketplace support. While the platform comprehensively manages digital interactions and data validation, it explicitly excludes the processing of in-app financial transactions, meaning all monetary exchanges must occur offline between the respective parties. Furthermore, the boundaries of the proposed solution do not include physical delivery logistics, biometric identity verification, or the provision of dedicated native iOS and Android applications, as the frontend is built entirely as a responsive web application.

2.6 Modules

The proposed system comprises the following primary functional modules:

2.6.1 Module 1: User Authentication & Account Management

- FE-1: Register and login securely using Email/Password with OTP verification.
- FE-2: Authenticate instantly using Google OAuth 2.0 integration.
- FE-3: Manage active stateless sessions securely via JSON Web Tokens (JWT).

2.6.2 Module 2: Product Listings & Ad Management

- FE-1: Create new used-goods listings using category-specific dynamic forms for Mobiles, Laptops, and Furniture.
- FE-2: Auto-extract and populate metadata specifications using regex-based text analysis on listing titles.
- FE-3: Transition listing statuses through a complete lifecycle including Draft, Active, Hidden, and Sold.

2.6.3 Module 3: Price Prediction

- FE-1: Generate a baseline market price utilizing a pre-trained ML ensemble of XGBoost, LightGBM, Random Forest, and Gradient Boosting.
- FE-2: Cross-validate baseline predictions with live web scraping of OLX Pakistan via the Groq LLM.
- FE-3: Filter scraped pricing data using Interquartile Range (IQR) mathematics to discard extreme outliers and ensure accuracy.

2.6.4 Module 4: Recommendations

- FE-1: Track user behavioral metrics including views, clicks, and favorites across different listing categories. Logs interaction details to the **UserActivity** table using the **RecommendationEngine**, applying activity weights and time-decay filters to build a real-time interest profile.
- FE-2: Generate a personalized discovery feed by utilizing SentenceTransformer embeddings ranked via cosine similarity algorithms. Ranks listings by calculating the cosine similarity between the user interest profile and listing vectors using the SemanticEmbeddingService, blended with category matching and post recency.

2.6.5 Module 5: Analytical Dashboard

- FE-1: Provide administrators with Recharts-based visual analytics tracking platform engagement and total active users.
- FE-2: Manage a centralized moderation queue allowing admins to manually review, approve, or reject flagged user listings.

2.6.6 Module 6: AR Customization Try-On

- FE-1: Convert standard 2D user-uploaded furniture images into photorealistic 3D GLB models using the Tripo AI API.
- FE-2: Render 3D furniture models directly within the user's physical space utilizing Google model-viewer for spatial WebAR.

2.6.7 Module 7: Fraud Prevention

- FE-1: Execute zero-shot CLIP classification on uploaded images to guarantee they match the declared listing category.
- FE-2: Identify and block duplicate or stolen photos across the platform utilizing dHash perceptual image hashing.

- FE-3: Scan listing descriptions using NLP algorithms to actively flag scam-baiting terminology before a listing is made public.

2.6.8 Module 8: Support & Notifications

- FE-1: Manage real-time system-wide alerts and visual unread count badges for all user activities.
- FE-2: Provide a dedicated support ticket dashboard allowing users to report issues and administrators to track resolution status.

2.6.9 Module 9: Messaging & Inbox

- FE-1: Facilitate secure, real-time WebSocket-based messaging between prospective buyers and sellers.
- FE-2: Organize conversations within an inbox view grouped by specific listing to streamline negotiations.

2.6.10 Module 10: Favorites & User Dashboard

- FE-1: Allow users to bookmark and save specific active listings into a dedicated favorites list for later viewing.
- FE-2: Provide a seller dashboard that tracks live status counts of the user's own active, pending, hidden, and sold advertisements.

2.6.11 Module 11: AI Chatbot Assistant

- FE-1: Provide an intelligent customer support and market guide powered by the Groq llama-3.1-8b-instant LLM.
- FE-2: Stream real-time conversational responses to the client using Server-Sent Events (SSE).
- FE-3: Inject live OLX CSV market context directly into the prompt to ensure the chatbot provides accurate platform queries.

3 Requirement Analysis

3.1 User classes and characteristics

Table 3: User Classes and Characteristics

User class	Description
Buyer	A registered individual looking to purchase second-hand mobiles, laptops, or furniture. They rely heavily on the system's AR visualization and semantic recommendations. Buyers interact with the system via the web platform to browse, favorite, and securely message sellers in real time without exposing personal phone numbers.
Seller	A registered individual looking to liquidate used electronics or furniture. They utilize the AI Price Predictor to set competitive prices and reduce the cognitive load of market research. Sellers interact with the system to create category-specific listings, manage ad statuses (Active, Hidden, Sold), and respond to buyer inquiries.
Administrator	A system moderator responsible for maintaining platform integrity. Administrators do not buy or sell; instead, they utilize the analytical dashboard to review fraud-flagged listings (via CLIP and dHash algorithms), manage user accounts, resolve support tickets, and track overall marketplace engagement metrics.

3.2 Requirement Identifying Technique

To translate user needs into functional system specifications, the following methodologies were employed:

- **Use Case Modeling**: Defined actor-system boundaries by mapping interactions between platform users (Buyers, Sellers, and Administrators) and the platform's core functional modules, detailing preconditions and successful outcomes.
- **Event-Response Tables**: Documented systemic triggers by mapping user events (e.g., uploading an image or inputting an outlier price) directly to backend AI processes (e.g., CLIP vision classification or price anomaly alerts) to establish predictable system feedback loops.
- **Storyboarding**: Charted sequential UI layouts to map the step-by-step user journey, visual feedback prompts, and state changes for the interactive AR Viewer and the floating glassmorphic Chatbot assistant.

3.2.1 Use case Diagram

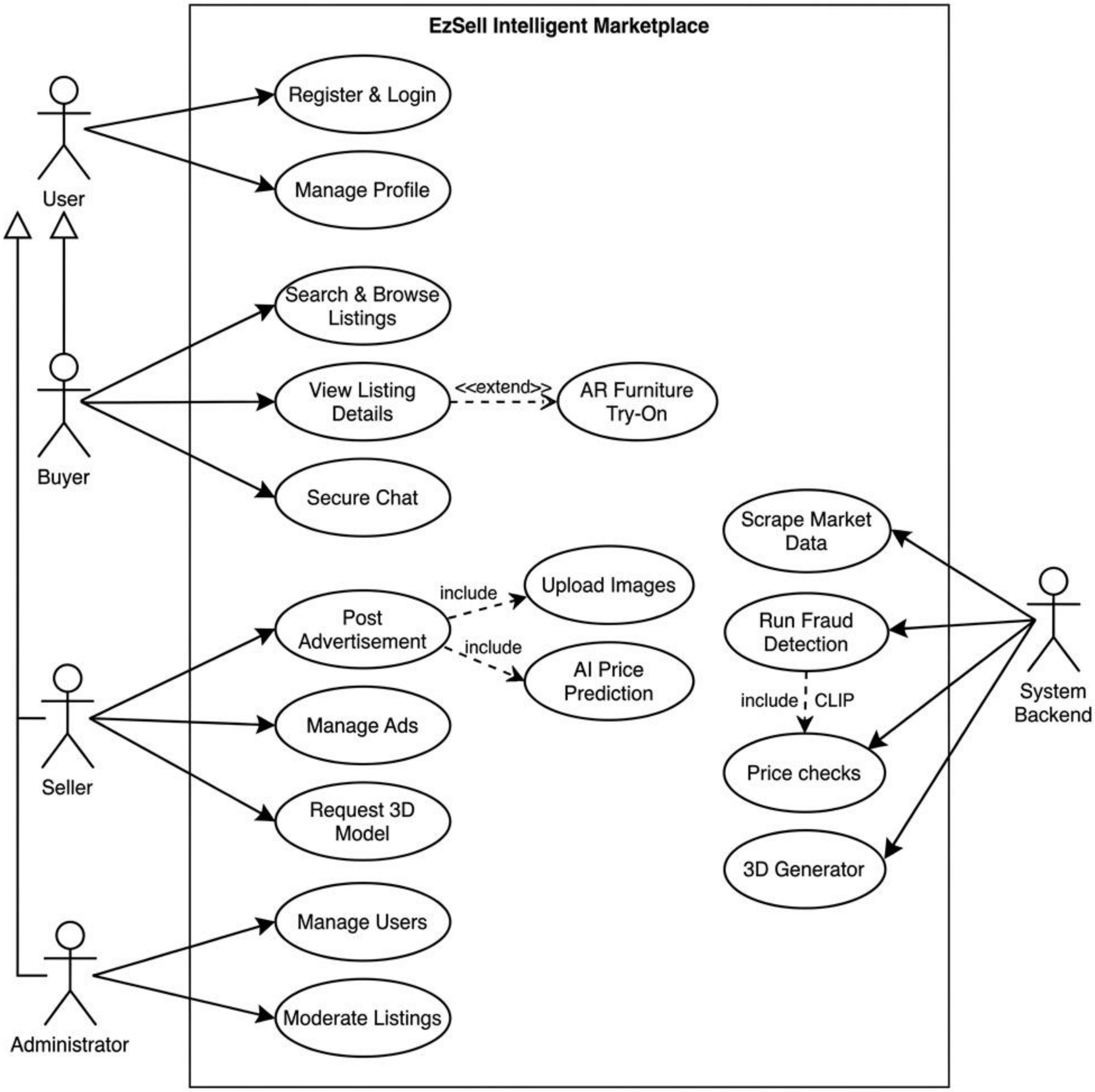

EzSell Intelligent Marketplace
User
Register & Login
Manage Profile
Buyer
Search & Browse Listings
View Listing Details
<<extend>>
AR Furniture Try-On
Secure Chat
Seller
Post Advertisement
include
Upload Images
include
AI Price Prediction
Manage Ads
Request 3D Model
Administrator
Manage Users
Moderate Listings
System Backend
Scrape Market Data
Run Fraud Detection
include
CLIP
Price checks
3D Generator

Figure : UML Use Case Diagram for the EzSell Intelligent Marketplace illustrating actor boundary interactions, generalized role inheritances (Buyer, Seller, and Administrator deriving from the base User), and direct integration points with the automated price prediction, fraud protection, and 3D AR visualization engines.

3.2.2 Detailed Use case

3.2.2.1 Use Case 01: Post a Product for Sale

Field	Description
Use Case ID	UC-01
Use Case Name	Post a Product for Sale
Primary Actor	Registered Seller
Goal	Allow users to list their pre-owned product with AI-extracted specs and market-based price suggestions.
Preconditions	User must be registered and logged in. Seller account verified.
Postconditions	Product listed and undergoing fraud detection.
Main Flow	1. Seller clicks 'Post New Product'. 2. Seller enters product title (e.g., 'Samsung S24 Ultra'). 3. System uses LLM to validate title and extract specs (RAM, Storage, Color). 4. System scrapes real-time market data from GSMArena and local Pakistani sites. 5. System suggests a price range based on condition selections. 6. Seller uploads images and confirms details. 7. System runs automated fraud detection (CLIP/dHash). 8. Product is listed and visible to buyers.
Alternate Flows	3a. If title is vague (e.g., 'Phone'), LLM provides specific improvement hints.
Exceptions	Image format unsupported -> show error; Incomplete fields -> prompt user.
Priority	High

3.2.2.2 Use Case 02: Search and Purchase a Product

Field	Description
Use Case ID	UC-02
Use Case Name	Search and Purchase a Product
Primary Actor	Registered Buyer
Goal	Enable buyers to find specific products using advanced filters and initiate purchase.
Preconditions	Internet connection active. User logged in.
Postconditions	Product marked 'In Negotiation' or 'Sold'.
Main Flow	1. Buyer enters keywords or uses category filters (Mobile, Laptop, Furniture). 2. System applies location-based and category-specific filters. 3. Buyer views results with thumbnails and engagement scores. 4. Buyer selects a product to view detailed specs and AR preview (if furniture). 5. Buyer clicks 'Contact Seller' to initiate chat. 6. Parties negotiate price and delivery terms. 7. Buyer confirms purchase; product status updated.
Alternate Flows	4a. AR try-on available for furniture -> buyer launches AR preview.
Exceptions	No products found -> show alternative recommendations.
Priority	High

3.2.2.3 *Use Case 03: Detect Fraudulent Listing*

Field	Description
Use Case ID	UC-03
Use Case Name	Detect Fraudulent Listing
Primary Actor	System (Fraud Detection), Admin
Goal	Automatically detect and flag fraudulent or duplicate listings.
Preconditions	Listing submitted by seller.
Postconditions	Listing approved, flagged for review, or rejected.
Main Flow	1. Listing data sent to Fraud Detection service. 2. System analyzes text for suspicious keywords (e.g., 'free', 'advance payment'). 3. System runs CLIP to detect stock images or misleading visuals. 4. System runs dHash to identify duplicate listings across the platform. 5. If score > threshold, listing is approved automatically. 6. If suspicious, listing is flagged for Admin manual review.
Alternate Flows	6a. Admin reviews flagged post and takes final action (Ban/Approve).
Exceptions	ML Model timeout -> default to manual moderation.
Priority	Medium-High

3.2.2.4 Use Case 04: Visualize Furniture in AR

Field	Description
Use Case ID	UC-04
Use Case Name	Visualize Furniture in AR
Primary Actor	Registered Buyer
Goal	Allow buyers to see how a furniture item fits in their actual physical environment.
Preconditions	Device supports WebXR. Product has a 3D model.
Postconditions	Buyer gains confidence in fit and style.
Main Flow	1. Buyer clicks 'View in AR' on a furniture product page. 2. System launches AR viewer and requests camera access. 3. Buyer scans floor/walls to establish grounding. 4. 3D model appears anchored to the surface. 5. Buyer moves, rotates, and snaps the model to wall axes. 6. System provides haptic feedback for alignment. 7. Buyer takes screenshot or proceeds to contact seller.
Alternate Flows	4a. No 3D model exists -> System prompts to generate one from images.
Exceptions	AR hardware incompatible -> show 3D web viewer fallback.
Priority	High

3.2.2.5 Use Case 05: Secure In-App Messaging

Field	Description
Use Case ID	UC-05
Use Case Name	Secure In-App Messaging
Primary Actor	Buyer, Seller
Goal	Facilitate real-time negotiation and communication between parties.
Preconditions	Buyer initiates contact from listing page.
Postconditions	Chat history preserved for safety and reference.
Main Flow	1. Buyer sends a message to the seller. 2. System delivers real-time notification to the seller. 3. Seller opens Chat Window and responds. 4. Parties exchange details, images, and price offers. 5. System monitors for prohibited keywords (fraud prevention). 6. Negotiation concludes with an agreement or cancellation.
Alternate Flows	5a. Automated chatbot handles common inquiries if seller is offline.
Exceptions	Socket connection lost -> show retry indicator.
Priority	High

3.2.2.6 Use Case 06: Market Price Suggestion Service

Field	Description
Use Case ID	UC-06
Use Case Name	Market Price Suggestion Service
Primary Actor	System (Pricing Engine)
Goal	Ground price suggestions in actual market data to prevent over/underpricing.
Preconditions	Product title and category provided.
Postconditions	Fair market value range generated.
Main Flow	1. System triggers scraping of GSMArena (mobiles) or local tech sites. 2. System gathers multiple price snippets from search results. 3. System applies IQR (Interquartile Range) filtering to remove outliers. 4. LLM analyzes filtered data to determine 'New' and 'Used' market prices. 5. System factors in user-selected condition and specs. 6. Final suggested range displayed to seller.
Alternate Flows	3a. No data found -> default to LLM general knowledge.
Exceptions	Rate limit hit -> use cached price data.
Priority	High

3.2.2.7 Use Case 07: Personalized Feed Generation

Field	Description
Use Case ID	UC-07
Use Case Name	Personalized Feed Generation
Primary Actor	System (Recommendation Service)
Goal	Increase engagement by showing relevant products to users based on behavior.
Preconditions	User has browsing/engagement history.
Postconditions	Personalized 'For You' section updated.
Main Flow	1. System tracks user clicks, views, and favorites. 2. Engagement scores are calculated for categories and brands. 3. Recommendation engine fetches listings matching high-score attributes. 4. Feed is sorted by relevance and recency. 5. Feed is served to the homepage dashboard.
Alternate Flows	3a. New user -> show trending/popular products.
Exceptions	Profile data missing -> show general top listings.
Priority	Medium

3.2.2.8 *Use Case UC-08: Automated 3D Model Generation*

Field	Description
Use Case ID	UC-08
Use Case Name	Automated 3D Model Generation
Primary Actor	Registered Seller
Goal	Transform 2D images into interactive 3D models for furniture listings.
Preconditions	Seller uploads high-quality product images.
Postconditions	3D model (.glb) generated and attached to listing.
Main Flow	1. Seller selects 'Generate 3D' option. 2. Images are sent to Tripo AI API via backend service. 3. System polls API for generation progress. 4. System downloads and optimizes the resulting mesh. 5. Model is stored in cloud storage and linked to listing. 6. Seller reviews model in 3D viewer.
Alternate Flows	4a. Generation quality low -> prompt for better lighting/angles.
Exceptions	API credits exhausted -> notify user of delay.
Priority	Medium-High

3.2.3 Event Response Table

Table 1: User Interface and Navigation Events

Event	System State	Response	Frequency
User clicks "Post Ad" button or uses Alt+P shortcut	User is logged in and authenticated. Main dashboard is displayed.	1. Verify user authentication status 2. Display "Create New Advertisement" form 3. Show category selection options 4. Initialize form validation system 4. Data Elements: User ID, Session Token Resulting State: Advertisement form displayed, awaiting category selection	High (Multiple per session)
User selects product category (Electronics/Furniture)	Advertisement form is displayed with category options available.	1. Load category-specific specification fields 2. Display relevant input forms (Electronics: Brand, Model, Storage; Furniture: Material, Dimensions) 3. Enable real-time validation for selected category Data Elements: Category Type, Field Templates Resulting State: Category-specific form displayed, ready for specification entry	Medium (Once per ad)
User enters product specifications and description	Category-specific form is displayed with empty specification fields.	1. Validate input format in realtime 2. Provide dropdown suggestions where applicable 3. Show tooltips and guidance messages 4. Prepare data for AI price prediction Data Elements: Product Title, Description, Brand, Model, Condition, Specifications Resulting State: Specifications	Medium (Once per ad)

Table 2: Image Upload and Media Management Events

Event	System State	Response	Frequency
User uploads product images	Product specifications are entered. Image upload section is available.	1. Validate image format (JPG, PNG, WebP) and size (<5MB) 2. Upload to cloud storage (AWS S3/Cloudinary) 3. Generate thumbnails for display 4. Run basic image analysis for fraud detection 5. Allow image reordering Data Elements: Image Files, File Metadata, Upload Progress Resulting State: Images uploaded and validated, ready for AI price prediction	Medium (1-5 per ad)
Image upload fails	User attempting to upload image. Network timeout, unsupported format, or file size exceeds limit.	1. Display specific error message (format, size, or network issue) 2. Provide options to retry, modify image, or continue 3. If minimum 1 image uploaded, allow continuation 4. Save form draft if user cancels Data Elements: Error Type, File Information, Draft Data Resulting State: Error displayed with recovery options, or draft saved	Low (5-10% of uploads)

Event	System State	Response	Frequency
	option.	3. Show warning if price significantly deviates 4. Allow override with additional verification if needed Data Elements: Custom Price, Validation Range, Override Request Resulting State: Custom price set, ready for contact details	pricing)
AI Price Prediction System unavailable	User requests price suggestion. AI system timeout or error occurs.	1. Display system unavailable message 2. Offer retry after 30 seconds 3. Provide alternative: market price range from recent listings 4. Allow user to continue without AI suggestion or save draft Data Elements: Error Code, Fallback Price Data, User Choice Resulting State: Fallback pricing provided or process saved as draft	Low (15% system downtime)

Table 3: AI Price Prediction and Pricing Events

Event	System State	Response	Frequency
User clicks "Get Price Suggestion"	Product specifications and images are uploaded. AI Price Prediction System is operational.	1. Send specifications to AI Price Prediction Module (M3) 2. Display loading indicator 3. Wait for AI response (max 1.5 seconds per PER-2) 4. Prepare for price display Data Elements: Product Specs, Market Data Request Resulting State: AI prediction request in progress	High (Once per ad)
AI Price Prediction System responds with suggestion	AI prediction request is in progress. System awaiting response within 1.5 seconds.	1. Display suggested price range with confidence level 2. Show market analysis and justification 3. Provide options to accept or set custom price 4. Log prediction data for system learning Data Elements: Price Range, Confidence Level, Market Analysis Resulting State: Price suggestion displayed, awaiting user decision	High (85% success rate)
User accepts AI price suggestion	AI price suggestion is displayed with confidence level and market analysis.	1. Set suggested price as final asking price 2. Validate price is within market range 3. Continue to contact information section 4. Update price acceptance statistics Data Elements: Accepted Price, User Choice Statistics Resulting State: Price confirmed, ready for contact details	High (70% acceptance)
User rejects AI suggestion and sets custom price	AI price suggestion is displayed. User chooses custom pricing	1. Display custom price input field 2. Validate against market extremes (10%-300% of AI suggestion)	Medium (30% custom

Table 4: Security and Fraud Detection Events

Event	System State	Response	Frequency
Fraud Detection System identifies suspicious patterns	User submits final advertisement. System runs fraud detection analysis.	1. Hold advertisement from going live 2. Display message: "Your listing is under review for quality assurance" 3. Flag for administrator review within 24 hours 4. Generate alert with suspicious pattern details Data Elements: Fraud Score, Pattern Details, Admin Alert Resulting State: Advertisement held in review queue	Low (5% of ads)
Administrator approves flagged advertisement	Advertisement is in review queue. Administrator completes manual review.	1. Change advertisement status to "Active" 2. Make advertisement live and searchable 3. Send approval notification to seller 4. Update fraud detection learning data Data Elements: Admin Decision, Approval Reason, Notification Data Resulting State: Advertisement live and active	Low (60% of flagged)
Administrator rejects flagged advertisement	Advertisement is in review queue. Administrator identifies policy violation.	1. Keep advertisement status as "Rejected" 2. Send detailed rejection explanation to seller 3. Provide opportunity to revise and resubmit 4. Update user violation history Data Elements: Rejection Reason, Revision Guidelines, User History Resulting State: Advertisement rejected, seller notified with revision option	Low (40% of flagged)

Table 5: System Events and Error Handling

Event	**System State**	**Response**	**Frequency**
User session expires during advertisement posting	User is in middle of posting process. Session timeout occurs.	1. Save current form data as draft 2. Redirect to login page with recovery message 3. Upon re-authentication, offer to restore saved data 4. Continue from last completed step if restored Data Elements: Draft Data, Session Info, Recovery Token Resulting State: User redirected to login with recovery option	Medium (10-15% of sessions)
System validates invalid input data	User submits form with invalid data (price out of range, missing required fields).	1. Highlight invalid fields with specific error messages 2. Prevent form submission until corrected 3. Provide guidance for acceptable values 4. Maintain other valid data in form Data Elements: Validation Errors, Field Values, Guidance Text Resulting State: Form displayed with validation errors, awaiting correction	Medium (20-30% of submissions)
Advertisement posting completes successfully	All validations pass. Advertisement ready to go live.	1. Generate unique advertisement ID 2. Store advertisement in database with "Active" status 3. Make advertisement searchable by buyers 4. Send confirmation email/SMS to seller 5. Schedule automatic expiration based on duration 6. Display success confirmation with ID Data Elements: Ad ID, Confirmation Details, Expiration Schedule Resulting State: Advertisement live and active, seller notified	High (85-90% success)

3.3 Functional Requirements

Table 4 Description of FR-1

Identifier	FR-1
Title	Profile Management
Requirement	The system shall allow users to register securely via Email or Google OAuth, authenticate via JWT, and manage their profiles.
Source	User Requirement
Rationale	To securely onboard users and manage identity.
Business Rule	Passwords must be hashed via bcrypt.
Dependencies	None
Priority	High

Table 5 Description of FR-2

Identifier	FR-2
Title	Product Listings & Management
Requirement	The system shall enable sellers to create, edit, hide, and delete category-specific listings with multi-image uploads and AI title validation.
Source	Core Marketplace
Rationale	To populate the platform with structured inventory.
Business Rule	Must select Mobile, Laptop, or Furniture.
Dependencies	FR-1
Priority	High

Table 6 Description of FR-3

Identifier	FR-3
Title	Price Prediction
Requirement	The system shall calculate a fair market price utilizing a blended ML ensemble (XGBoost, LightGBM, RF, GBC) and live Groq LLM web scraping (Accuracy: 85%+).
Source	AI Engine
Rationale	To reduce seller cognitive load and ensure fair pricing.
Business Rule	Predictions must be cross-validated against live scraped data.
Dependencies	FR-2
Priority	High

Table 7 Description of FR-4

Identifier	FR-4
Title	Recommendations
Requirement	The system shall generate a personalized discovery feed by encoding listings with SentenceTransformer vectors and ranking via cosine similarity.
Source	User Experience
Rationale	To reduce search fatigue and match user intent.
Business Rule	Cosine similarity > 0.5 threshold.
Dependencies	FR-2
Priority	Medium

Table 8 Description of FR-5

Identifier	FR-5
Title	Analytical Dashboard
Requirement	The system shall provide administrators with visual Recharts tracking engagement, listings, users, and support tickets for platform moderation.
Source	Admin Team
Rationale	To oversee and moderate marketplace health.
Business Rule	Requires Admin privileges.
Dependencies	FR-1
Priority	High

Table 9 Description of FR-6

Identifier	FR-6
Title	AR Customization Try-On
Requirement	The system shall invoke the Tripo AI API to generate photorealistic 3D GLB models from images and render them in physical space via WebAR.
Source	AR Engine
Rationale	To eliminate spatial anxiety for furniture buyers.
Business Rule	Restricted strictly to the Furniture category.
Dependencies	FR-2
Priority	High

Table 10 Description of FR-7

Identifier	FR-7
Title	Fraud Prevention
Requirement	The system shall utilize CLIP zero-shot classification, dHash deduplication, and NLP to proactively block duplicate, stolen, or scam listings.
Source	Security Team
Rationale	To protect marketplace integrity and prevent scams.
Business Rule	Hashes matched against global database.
Dependencies	FR-2
Priority	High

Table 11 Description of FR-8

Identifier	FR-8
Title	Support & Notifications
Requirement	The system shall facilitate a real-time ticketing dashboard and WebSocket-based notification alerts to keep users informed of status changes.
Source	Support Team
Rationale	To manage customer service workflows.
Business Rule	Real-time delivery required.
Dependencies	FR-1
Priority	Medium

Table 12 Description of FR-9

Identifier	FR-9
Title	Messaging & Inbox
Requirement	The system shall provide secure, real-time WebSocket messaging grouped by listing, allowing buyers and sellers to negotiate securely.
Source	Core Marketplace
Rationale	To facilitate instant communication without exposing emails.
Business Rule	Requires active session.
Dependencies	FR-1
Priority	High

Table 13 Description of FR-10

Identifier	FR-10
Title	Favorites & User Dashboard
Requirement	The system shall allow users to bookmark listings and provide sellers with a dashboard to track their active, pending, hidden, and sold items.
Source	User Experience
Rationale	To manage personal inventory and interests.
Business Rule	None
Dependencies	FR-2
Priority	Medium

Table 14 Description of FR-11

Identifier	FR-11
Title	AI Chatbot Assistant
Requirement	The system shall provide a conversational interface powered by llama-3.1-8b with live context injection to assist users via SSE streaming.
Source	AI Engine
Rationale	To provide automated intelligent support.
Business Rule	Market data context updated daily.
Dependencies	None
Priority	Medium

3.4 Non-Functional Requirements

This section specifies nonfunctional requirements defining system quality attributes. These requirements are specific, quantitative, and verifiable.

3.4.1 Reliability

- REL-1: The ML Pricing Engine shall maintain a Mean Time Between Failures (MTBF) of 720 hours. A failure is defined as an inability to generate a price estimate within 10 seconds. In the event of external API rate-limiting, the system shall implement automatic error detection and seamlessly fall back to local cached CSV datasets to prevent service disruption.

3.4.2 Usability

- USE-1: The platform shall allow a user to navigate from the homepage to a 3D AR furniture view with a maximum of three interactions (clicks/taps).
- USE-2: The listing creation form shall implement real-time inline validation to prevent submission errors, guiding the user to intuitively correct missing category-specific attributes.

3.4.3 Performance

- PER-1: 95% of personalized recommendation feeds generated by the SentenceTransformer model shall load completely within 800 milliseconds from the time of the user request over a standard 4G connection.
- PER-2: The 3D GLB model generation process must initiate browser rendering within 45 seconds of the user invoking the AR Try-On feature.

3.4.4 Security

- SEC-1: The system shall protect all user personal identifiable information (PII) and chat logs against unauthorized access, requiring advanced administrative credentials and secure cryptographic verification to access the central database.
- SEC-2: The application shall prevent automated bot scraping of seller contact information, ensuring phone numbers are completely obscured from unauthenticated entities.

3.5 External Interface Requirements

This section provides information ensuring the system communicates properly with users and external software/hardware elements.

3.5.1 User Interfaces Requirements

- UI-1: The web application shall adhere strictly to the Shadcn UI product family style guidelines for all form controls, alerts, and typography.
- UI-2: The interface shall implement a fully responsive grid system to accommodate standard mobile (375px width) and desktop (1440px width) screen resolution constraints.
- UI-3: A fixed navigation bar shall appear on every screen, providing instant access to the user inbox, favorites, and a floating action button for the AI Chatbot.

3.5.2 Software Interfaces

SI-1: AI Inference Services

- SI-1.1: The backend FastAPI service shall transmit uploaded listing images to the OpenAI CLIP model through a programmatic REST interface for category validation prior to database insertion.

SI-2: Tripo AI 3D Generation

- SI-2.1: The AR Engine shall send 2D image binaries to the Tripo AI API over HTTP and receive processed GLB 3D model URLs for browser rendering.

3.5.3 Hardware Interfaces

- HI-1: The frontend Single Page Application (SPA) shall request permissions and interface directly with the user's mobile device camera hardware via WebXR and Google model-viewer to project AR models into the physical environment.

3.5.4 Communications Interfaces

- CI-1: The system shall send an automated transactional email containing a 6-digit OTP to the user via SMTP protocols to confirm account activation.
- CI-2: The real-time messaging module shall utilize secure WebSocket (WSS) network protocols to transmit chat packets instantly between buyers and sellers.

4 Design and Architecture

4.1 Architectural Design

The system follows a Layered Architecture pattern with three main layers:

- **Presentation Layer:** User interface components including login and profile pages and product listing screens.
- **Business Logic Layer:** Core processing modules such as AI-based price prediction, analytics dashboard and AR visualization.
- **Data Layer:** Databases and storage systems that securely store users, listings, images, pricing models and activity logs.

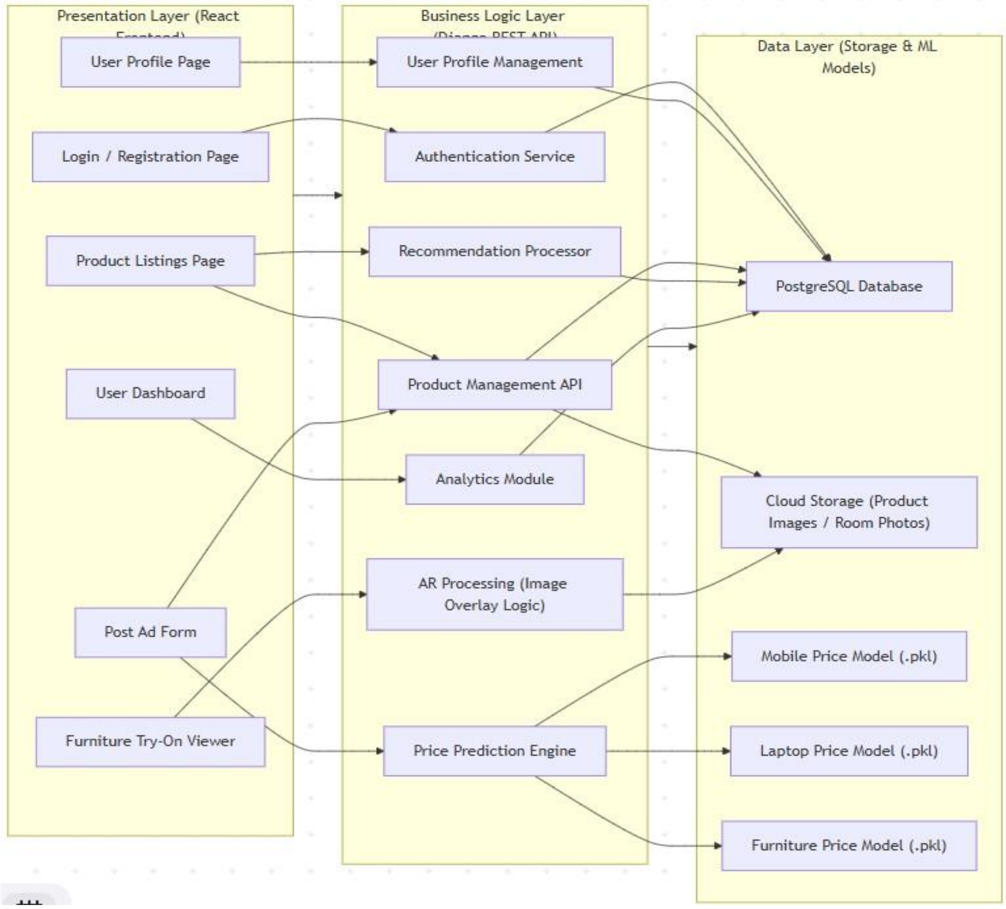

Presentation Layer (React
User Profile Page
Login / Registration Page
Product Listings Page
User Dashboard
Post Ad Form
Furniture Try-On Viewer
Business Logic Layer
User Profile Management
Authentication Service
Recommendation Processor
Product Management API
Analytics Module
AR Processing (Image Overlay Logic)
Price Prediction Engine
Data Layer (Storage & ML Models)
PostgreSQL Database
Cloud Storage (Product Images / Room Photos)
Mobile Price Model (.pkl)
Laptop Price Model (.pkl)
Furniture Price Model (.pkl)

Figure 1 Three-Tier Layered Architecture of EzSell

4.1.1 Activity Diagram:

4.1.1.1 UC-1 Profile Management Module

The user enters login details, the system checks the credentials, and if valid, the user is allowed into the system. If not, an error message is shown.

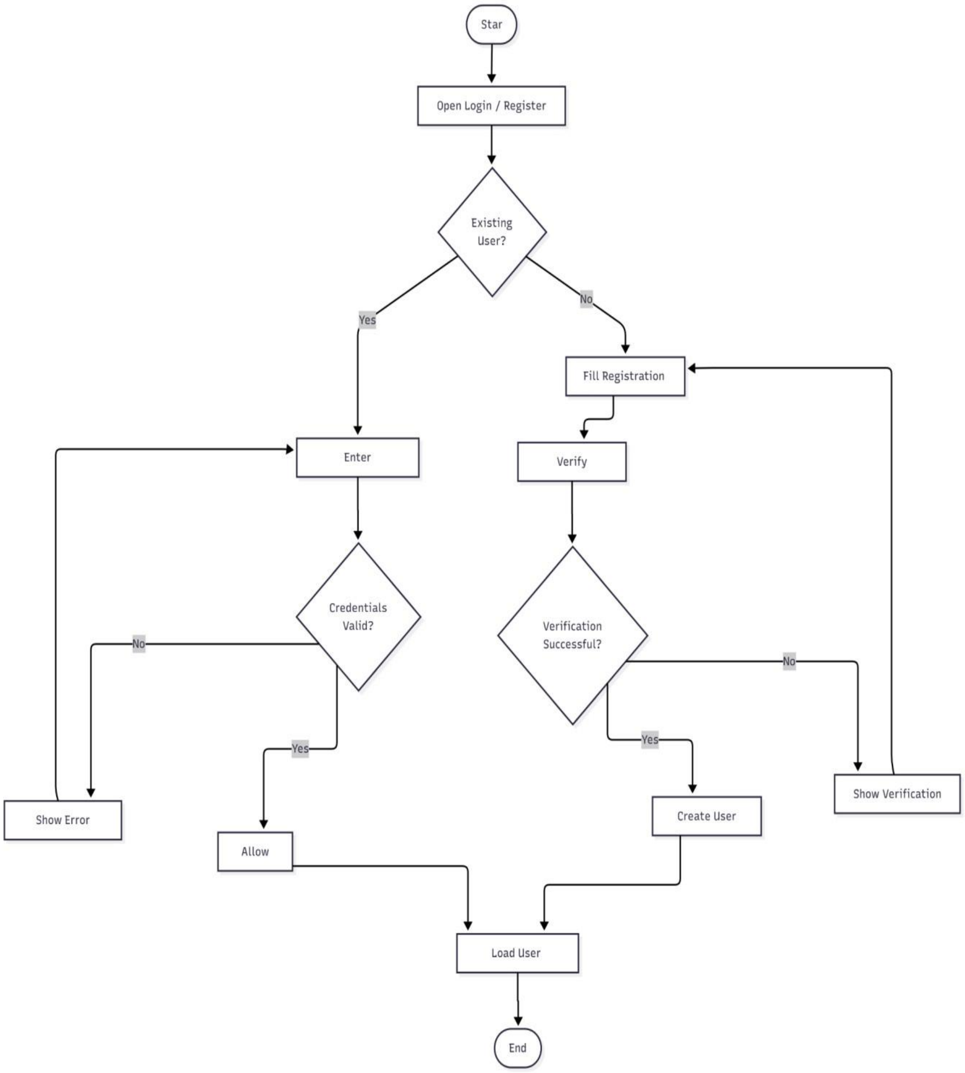

Star
Open Login / Register
Existing User?
Yes
No
Fill Registration
Enter
Verify
Credentials Valid?
No
Verification Successful?
No
Yes
Yes
Show Verification
Create User
Show Error
Allow
Load User
End

Figure 2 Activity Diagram — UC-1 Profile Management Module

4.1.1.2 UC-2 Product Listings and Management Module

The seller creates or edits a product listing, the system validates item details, uploads images, applies business rules, and stores the finalized listing in the database.

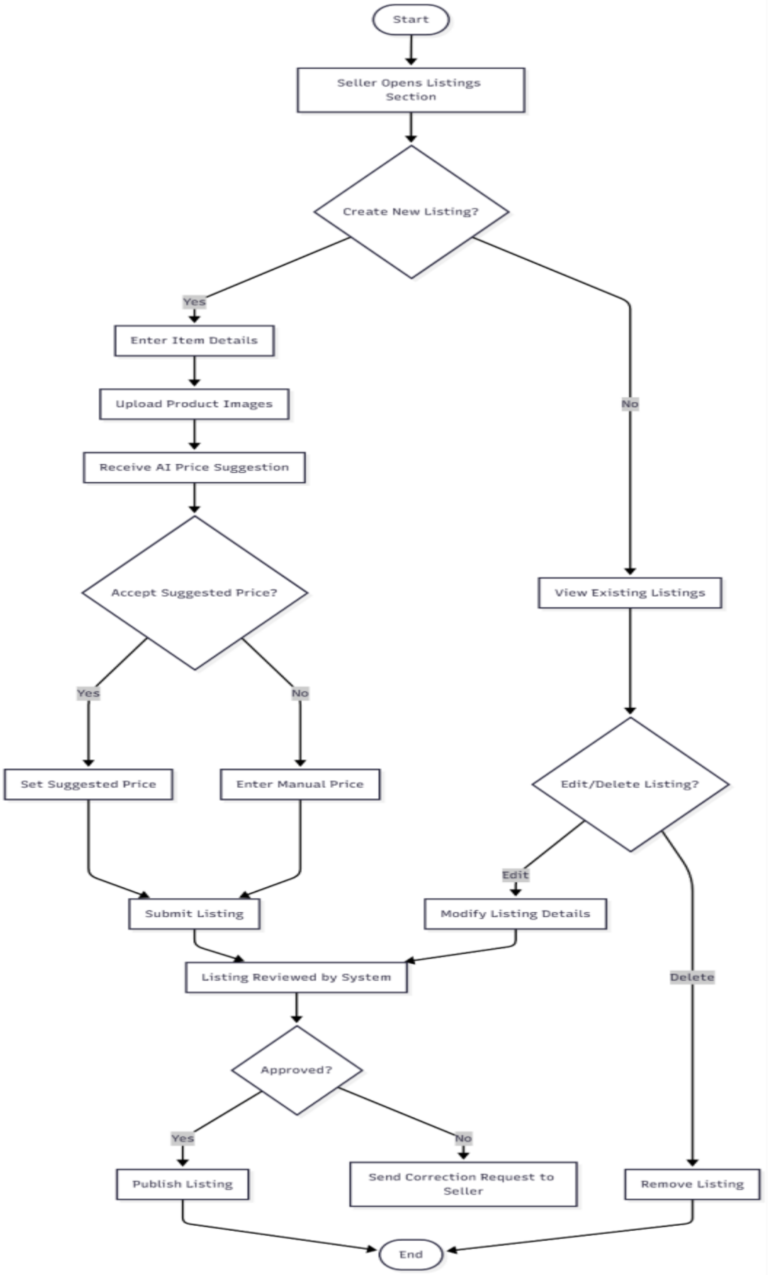

Start
Seller Opens Listings Section
Create New Listing?
Yes
No
Enter Item Details
Upload Product Images
Receive AI Price Suggestion
Accept Suggested Price?
Yes
No
Set Suggested Price
Enter Manual Price
View Existing Listings
Edit/Delete Listing?
Edit
Delete
Submit Listing
Modify Listing Details
Listing Reviewed by System
Approved?
Yes
No
Publish Listing
Send Correction Request to Seller
Remove Listing
End

Figure 3 Activity Diagram — UC-2 Product Listings and Management Module

4.1.1.3 UC-3 Price Prediction Module

The seller enters product specifications, the system processes the attributes, runs the AI pricing model, and returns a fair price suggestion before saving the listing.

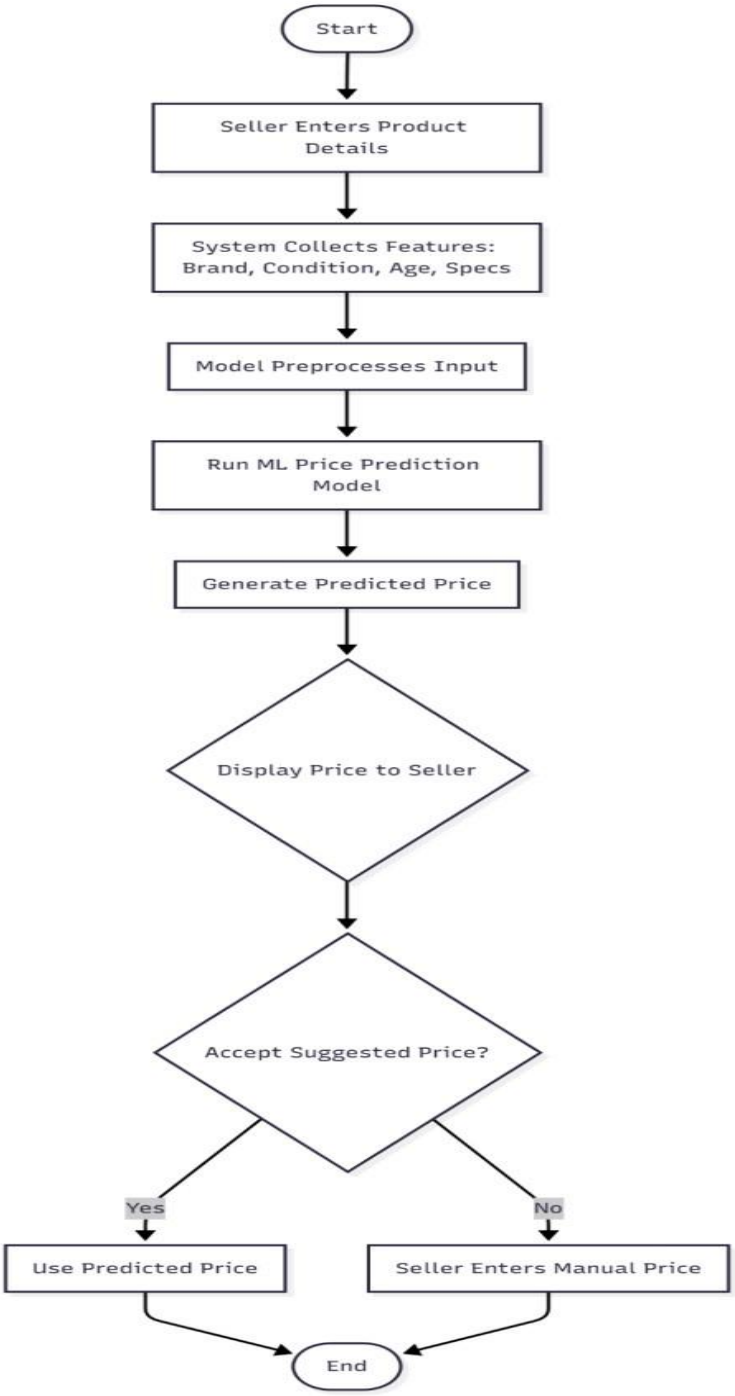

Start
Seller Enters Product Details
System Collects Features: Brand, Condition, Age, Specs
Model Preprocesses Input
Run ML Price Prediction Model
Generate Predicted Price
Display Price to Seller
Accept Suggested Price?
Yes
No
Use Predicted Price
Seller Enters Manual Price
End

Figure 4 Activity Diagram — UC-3 Price Prediction Module

4.1.1.4 UC-4 Analytical Dashboard

The system collects user activity and listing data, processes trends and metrics, and displays insights such as listing performance and engagement analytics on the dashboard.

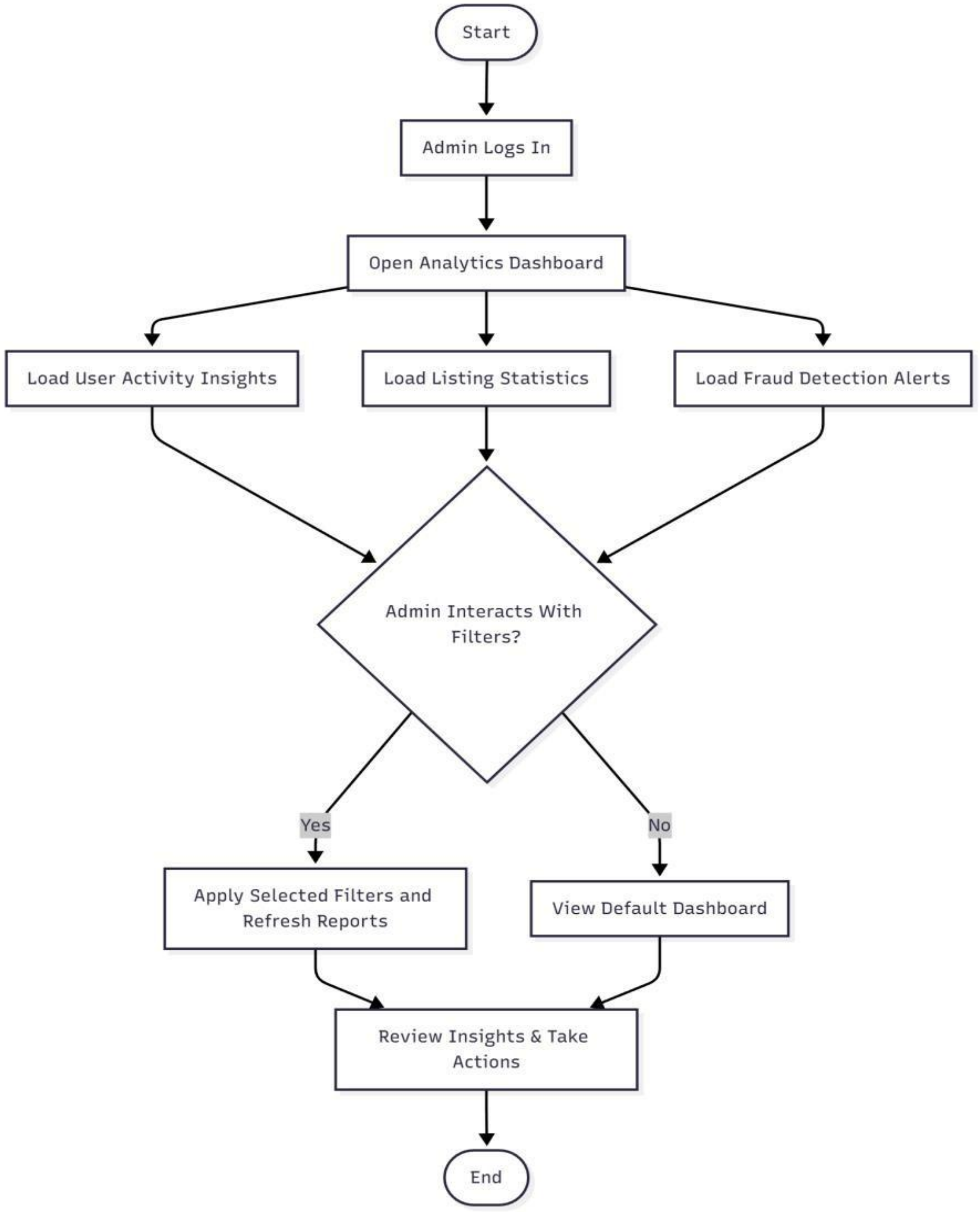

Start
Admin Logs In
Open Analytics Dashboard
Load User Activity Insights
Load Listing Statistics
Load Fraud Detection Alerts
Admin Interacts With Filters?
Yes
No
Apply Selected Filters and Refresh Reports
View Default Dashboard
Review Insights & Take Actions
End

Figure 5 Activity Diagram — UC-4 Analytical Dashboard Module

4.1.2 Class Diagram: ORM Schema

Represents SQLAlchemy ORM. Highlights polymorphic relationships between base Listing and specialized Mobile/Laptop/Furniture entities.

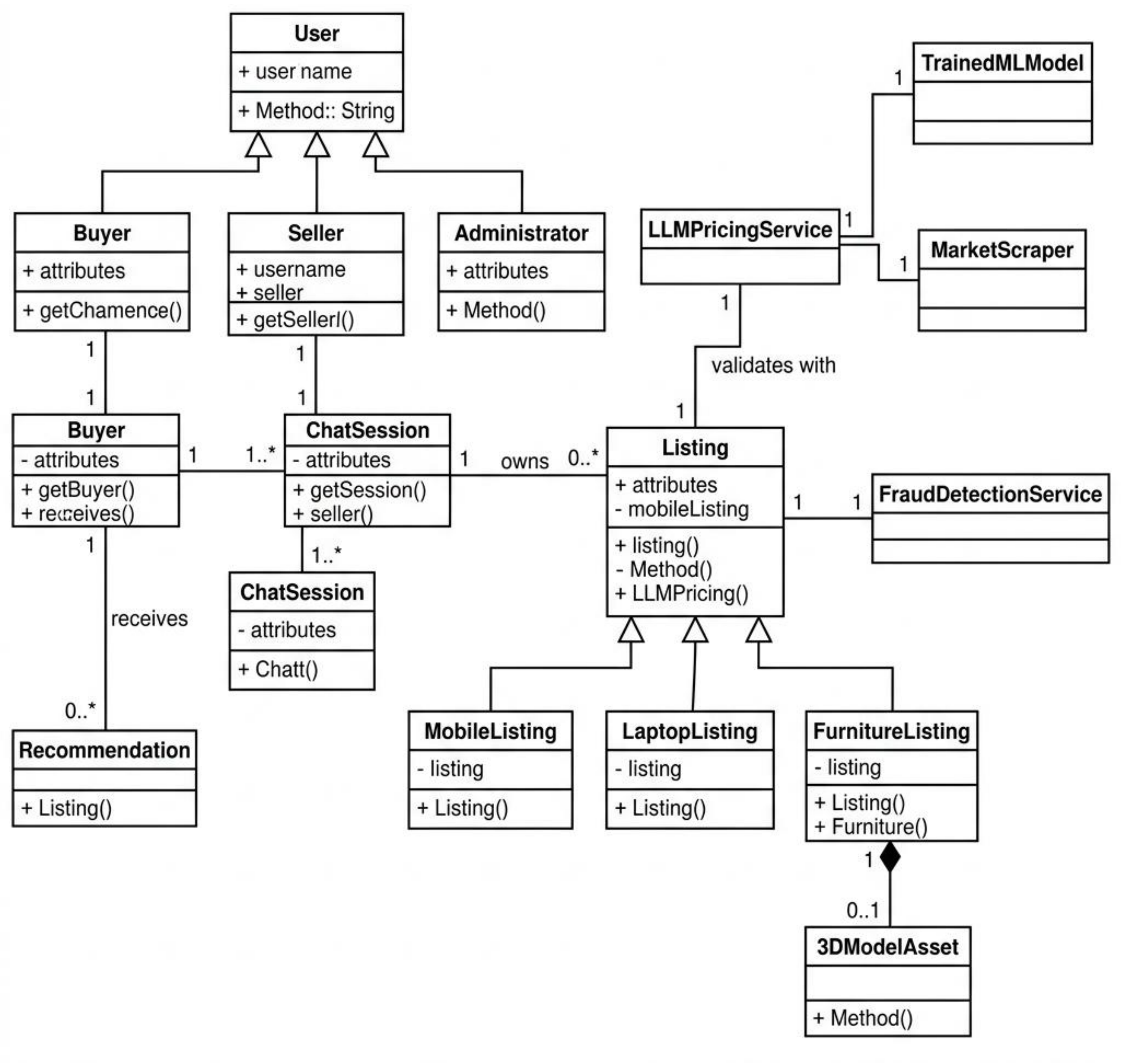

User
+ user name
+ Method:: String
Buyer
+ attributes
+ getChamence()
Seller
+ username
+ seller
+ getSellerI()
Administrator
+ attributes
+ Method()
TrainedMLModel
LLMPricingService
MarketScraper
1
1
1
1
1
1
1
1
validates with
1
Buyer
- attributes
+ getBuyer()
+ receives()
1
1..*
ChatSession
- attributes
+ getSession()
+ seller()
1
owns
0..*
Listing
+ attributes
- mobileListing
+ listing()
- Method()
+ LLMPricing()
1
1
FraudDetectionService
1
1..*
ChatSession
- attributes
+ Chatt()
receives
0..*
Recommendation
+ Listing()
MobileListing
- listing
+ Listing()
LaptopListing
- listing
+ Listing()
FurnitureListing
- listing
+ Listing()
+ Furniture()
1
0..1
3DModelAsset
+ Method()

Figure 6 UML Class Diagram for the EzSell Marketplace

4.1.3 System Sequence Diagram

4.1.3.1 UC-1 Profile Management

Shows how the user sends registration or login inputs to the system, how the system verifies credentials, and how access is granted along with profile data retrieval.

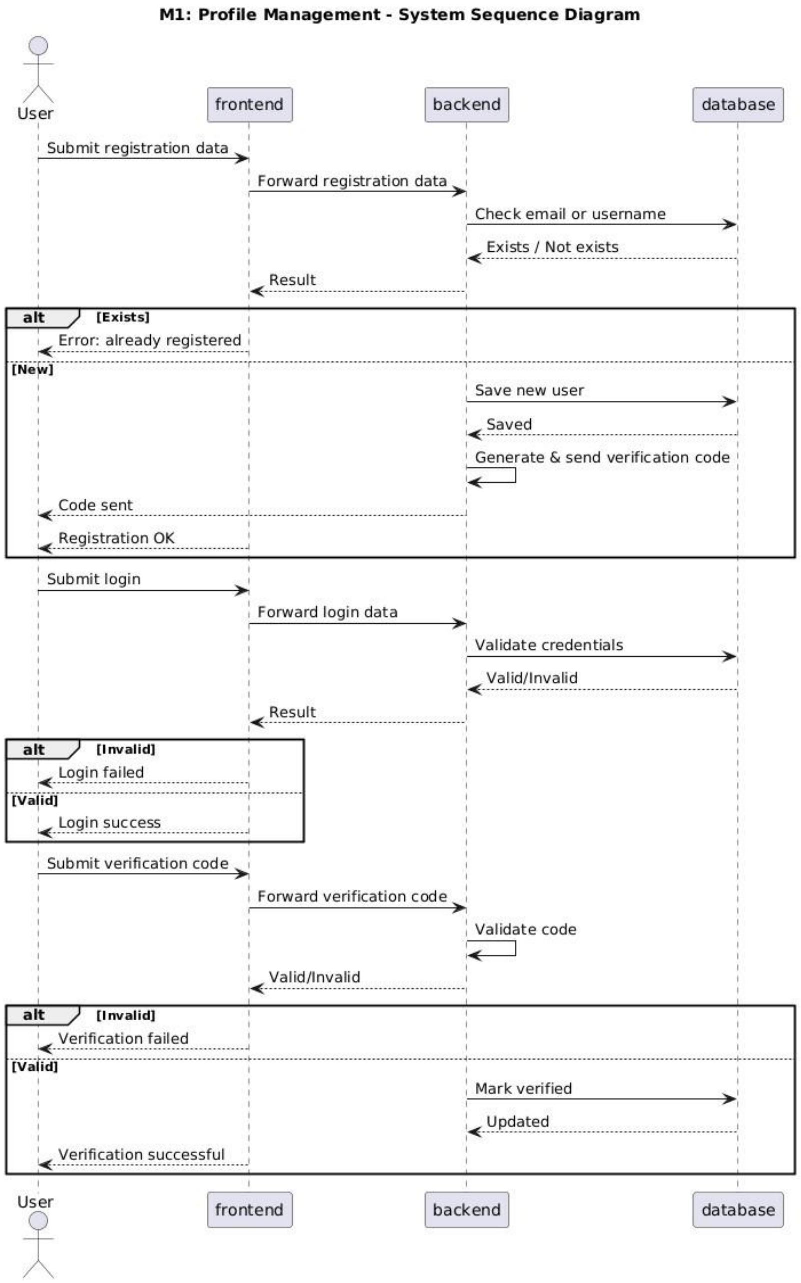

M1: Profile Management - System Sequence Diagram
User
frontend
backend
database
Submit registration data
Forward registration data
Check email or username
Exists / Not exists
Result
alt [Exists]
Error: already registered
[New]
Save new user
Saved
Generate & send verification code
Code sent
Registration OK
Submit login
Forward login data
Validate credentials
Valid/Invalid
Result
alt [Invalid]
Login failed
[Valid]
Login success
Submit verification code
Forward verification code
Validate code
Valid/Invalid
alt [Invalid]
Verification failed
[Valid]
Mark verified
Updated
Verification successful
User
frontend
backend
database

Figure 7 System Sequence Diagram — M1: Profile Management

4.1.3.2 4.3.2 UC-2 Product Listing & Management

Illustrates how the seller submits product details and images, how the system validates inputs, applies business rules, and stores the listing in the database.

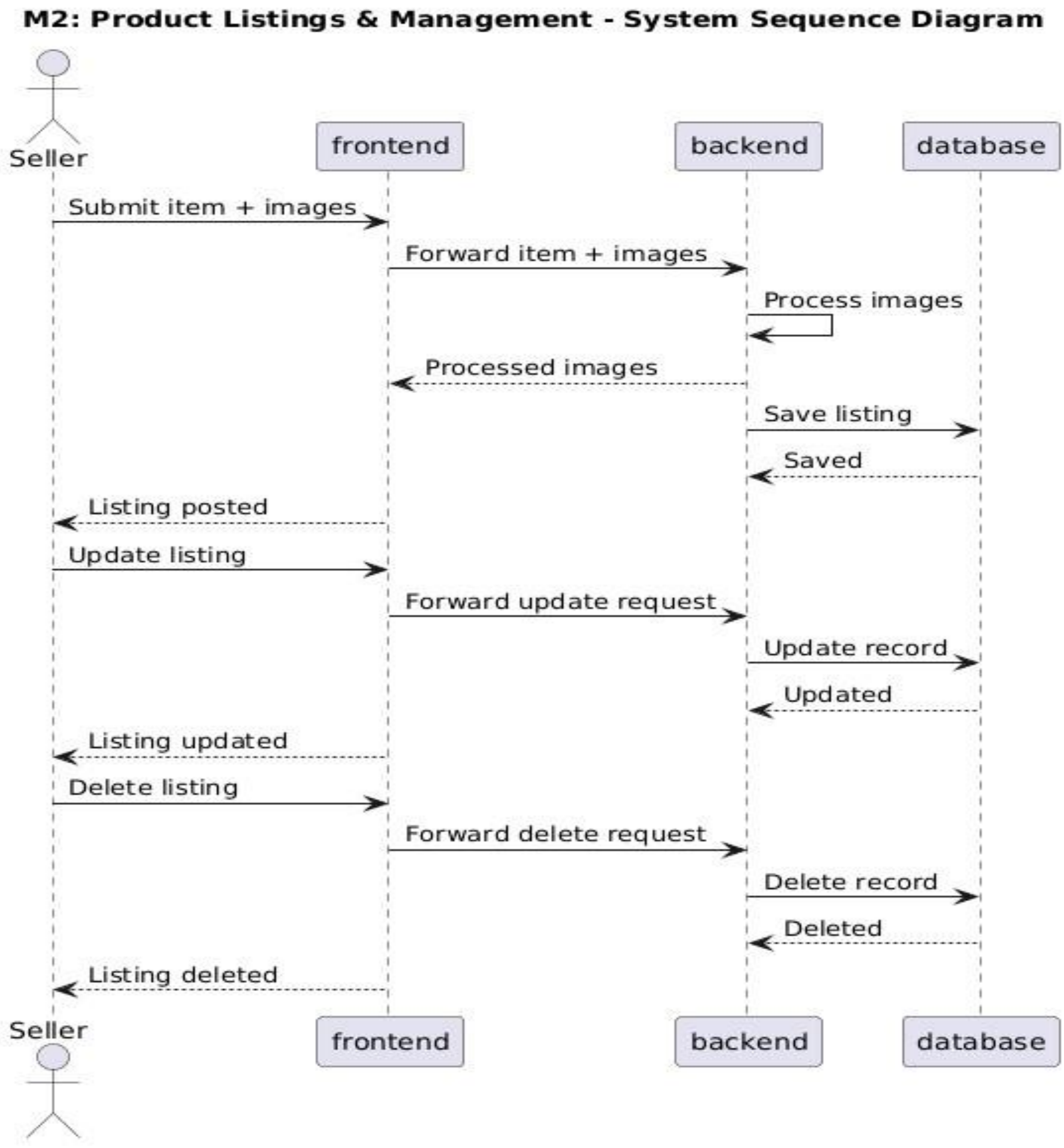

M2: Product Listings & Management - System Sequence Diagram
Seller
frontend
backend
database
Submit item + images
Forward item + images
Process images
Processed images
Save listing
Saved
Listing posted
Update listing
Forward update request
Update record
Updated
Listing updated
Delete listing
Forward delete request
Delete record
Deleted
Listing deleted
Seller
frontend
backend
database

Figure 8 System Sequence Diagram — M2: Product Listings and Management

4.1.3.3 UC -3 Price Prediction

Depicts how the seller provides product specifications, how the system sends the data to the AI pricing engine, receives a suggested price, and updates the listing before publishing.

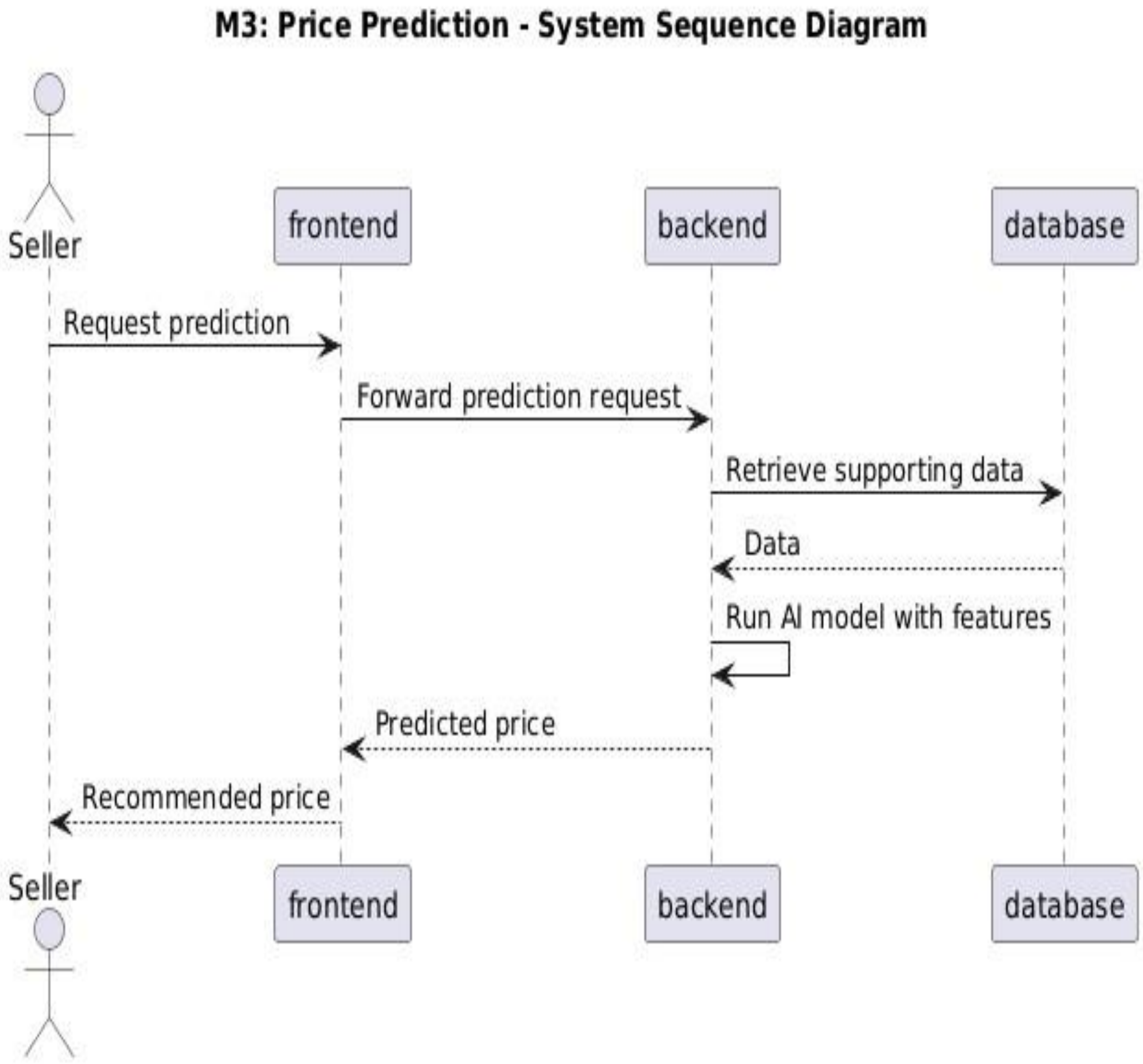

M3: Price Prediction - System Sequence Diagram
Seller
frontend
backend
database
Request prediction
Forward prediction request
Retrieve supporting data
Data
Run AI model with features
Predicted price
Recommended price
Seller
frontend
backend
database

Figure 9 System Sequence Diagram — M3: AI Price Prediction

4.1.3.4 UC-4 Analytical Dashboard

Shows how the system retrieves user and lists data, processes trends and metrics, and displays the insights on the dashboard for buyers, sellers, and admins.

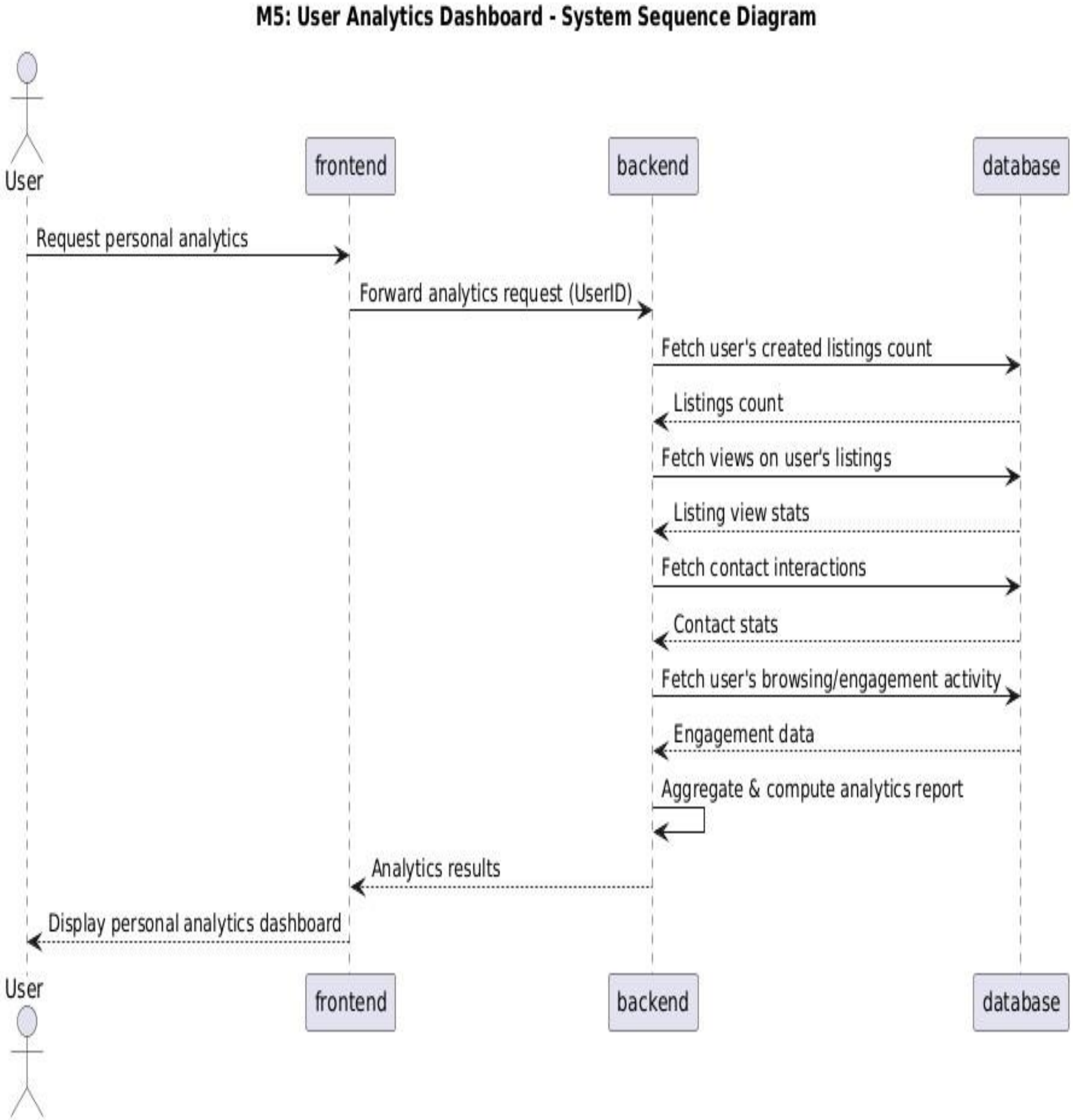

M5: User Analytics Dashboard - System Sequence Diagram
User
frontend
backend
database
Request personal analytics
Forward analytics request (UserID)
Fetch user's created listings count
Listings count
Fetch views on user's listings
Listing view stats
Fetch contact interactions
Contact stats
Fetch user's browsing/engagement activity
Engagement data
Aggregate & compute analytics report
Analytics results
Display personal analytics dashboard
User
frontend
backend
database

Figure 10 System Sequence Diagram — M5: User Analytics Dashboard

4.1.4 State Transition Diagram

4.1.4.1 4.4.1 UC-1 Profile Management

Shows how the system moves from “Logged Out” → “Validating Credentials” → “Logged In” if credentials are correct, or “Error” if validation fails.

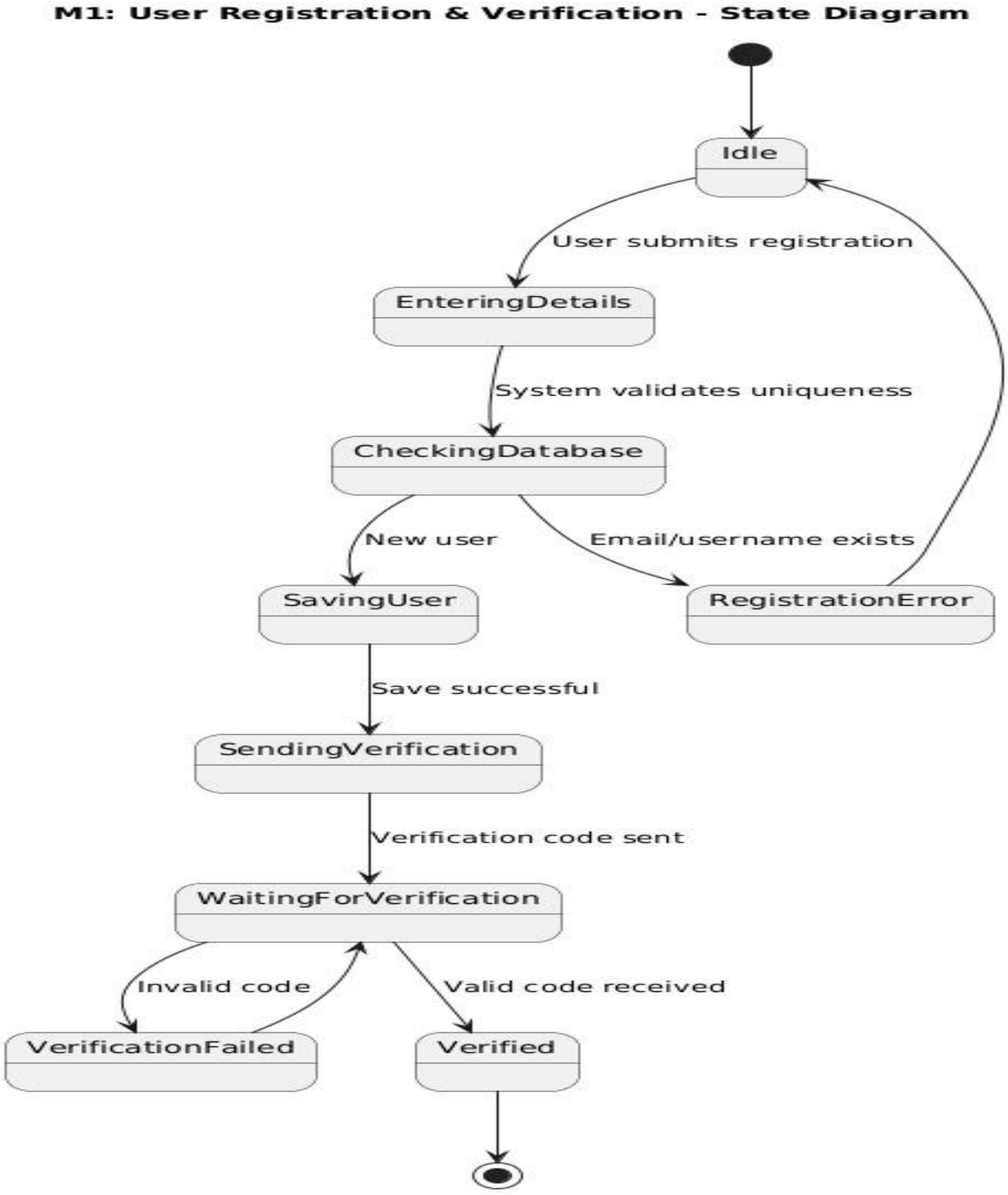

M1: User Registration & Verification - State Diagram
Idle
User submits registration
EnteringDetails
System validates uniqueness
CheckingDatabase
New user
Email/username exists
SavingUser
RegistrationError
Save successful
SendingVerification
Verification code sent
WaitingForVerification
Invalid code
Valid code received
VerificationFailed
Verified

Figure 11 State Transition Diagram — M1: User Registration and Verification

4.1.4.2 Product Listing & Management

Shows how a listing moves from “Draft” → “Validating Details” → “Published” if all inputs are correct, or “Error/Rejected” if validation fails.

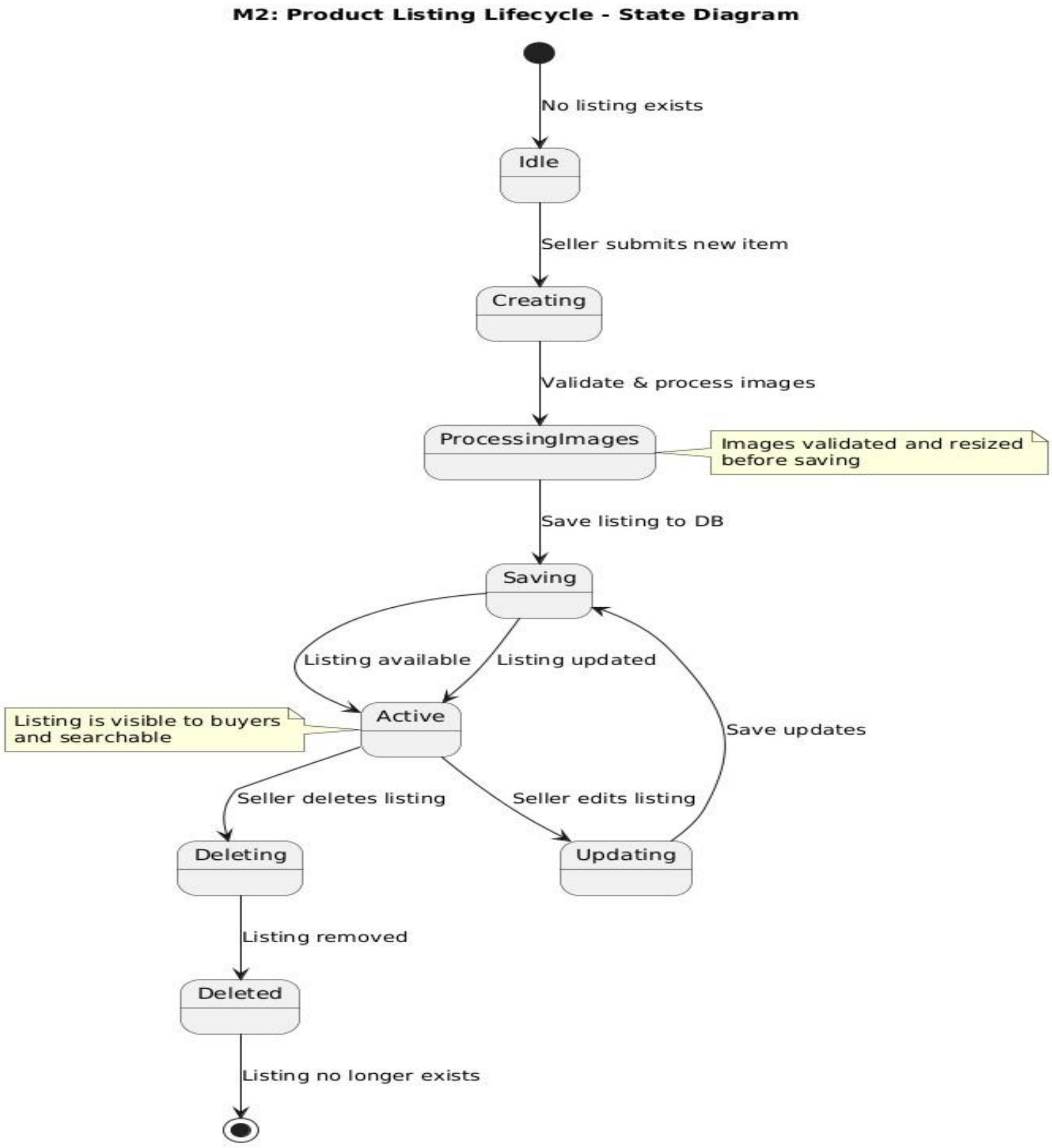
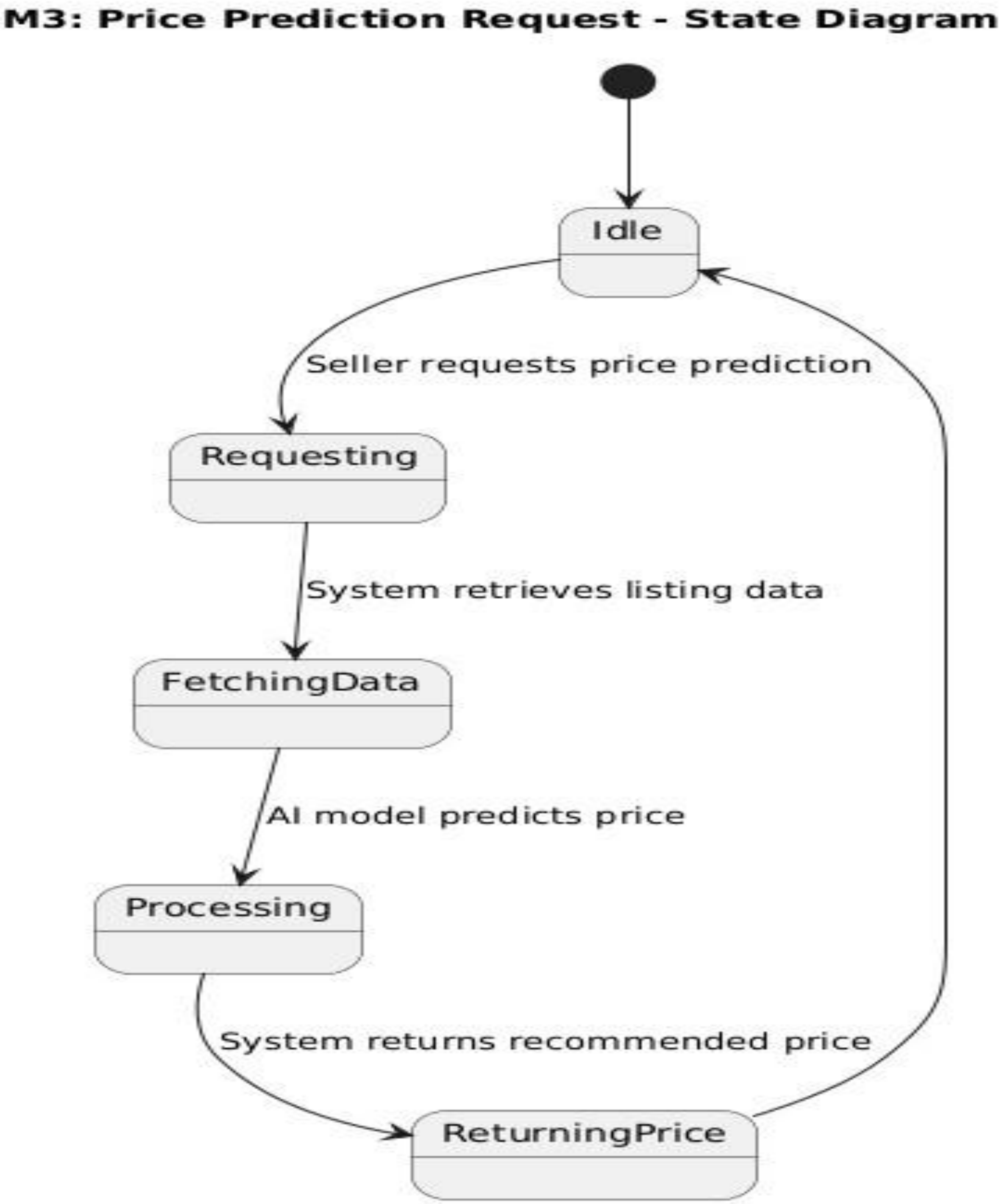

M2: Product Listing Lifecycle - State Diagram
No listing exists
Idle
Seller submits new item
Creating
Validate & process images
ProcessingImages
Images validated and resized before saving
Save listing to DB
Saving
Listing available
Listing updated
Save updates
Active
Listing is visible to buyers and searchable
Seller deletes listing
Seller edits listing
Deleting
Updating
Listing removed
Deleted
Listing no longer exists

Figure 12 State Transition Diagram — M2: Product Listing Lifecycle

4.1.4.3 Price Prediction

Shows how a product’s pricing moves from “Awaiting Price” → “Predicting Price” → “Price Suggested”, or “Manual Input Required” if AI cannot generate a valid suggestion.



M3: Price Prediction Request - State Diagram
Idle
Seller requests price prediction
Requesting
System retrieves listing data
FetchingData
AI model predicts price
Processing
System returns recommended price
ReturningPrice

Figure 13 State Transition Diagram — M3: Price Prediction Request

4.1.4.4 Analytical Dashboard

Shows how dashboard data moves from “Idle” → “Fetching Data” → “Processing Metrics” → “Displaying Insights”.

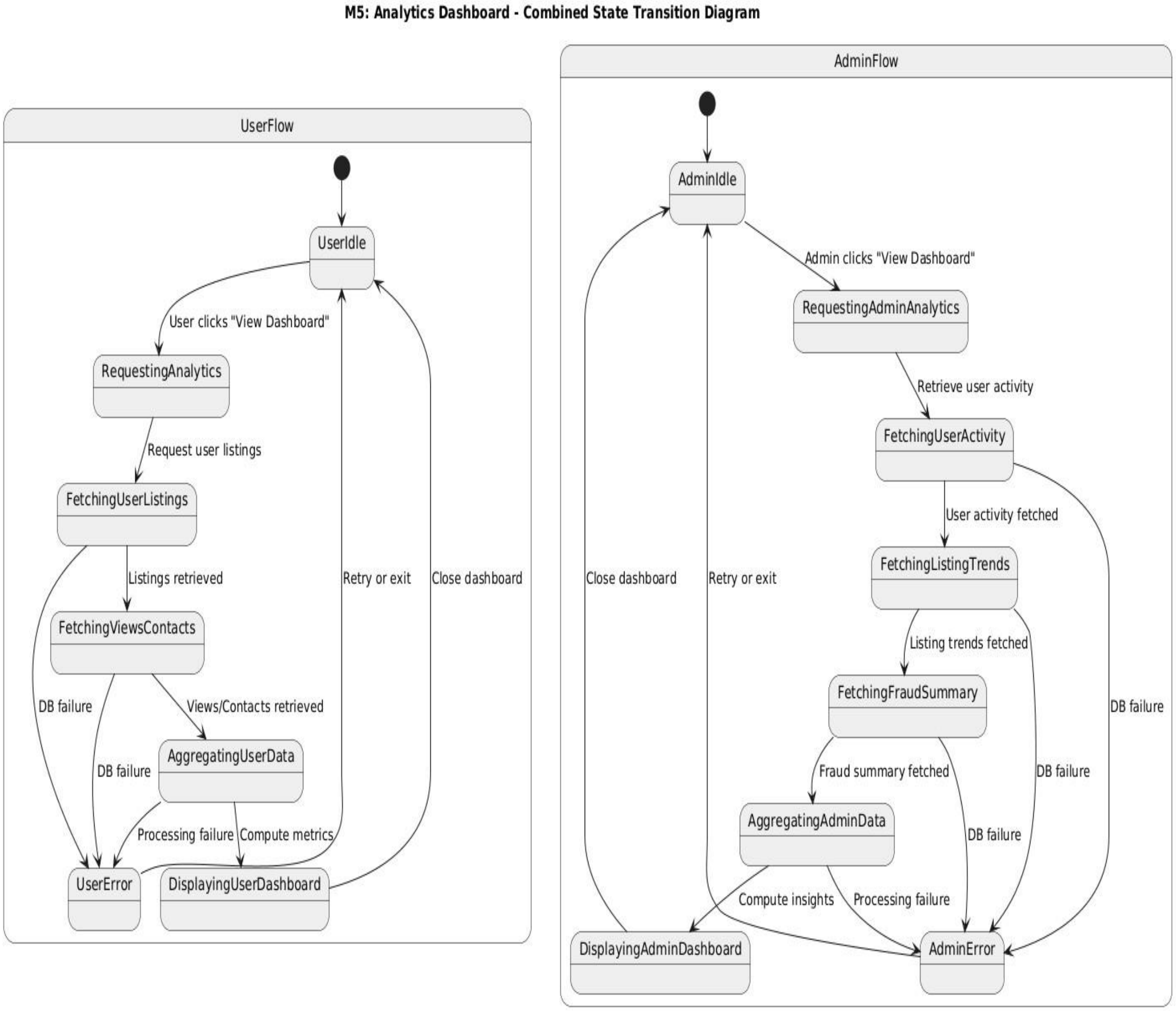

M5: Analytics Dashboard - Combined State Transition Diagram
UserFlow
UserIdle
User clicks "View Dashboard"
RequestingAnalytics
Request user listings
FetchingUserListings
Listings retrieved
FetchingViewsContacts
Views/Contacts retrieved
AggregatingUserData
DB failure
DB failure
Processing failure
Compute metrics
Retry or exit
Close dashboard
UserError
DisplayingUserDashboard
AdminFlow
AdminIdle
Admin clicks "View Dashboard"
RequestingAdminAnalytics
Retrieve user activity
FetchingUserActivity
User activity fetched
FetchingListingTrends
Listing trends fetched
FetchingFraudSummary
Fraud summary fetched
AggregatingAdminData
Compute insights
Processing failure
DB failure
DB failure
DB failure
Close dashboard
Retry or exit
DisplayingAdminDashboard
AdminError

Figure 14 Combined State Transition Diagram — M5: Analytics Dashboard

4.2 Data Design

4.2.1 Data Dictionary

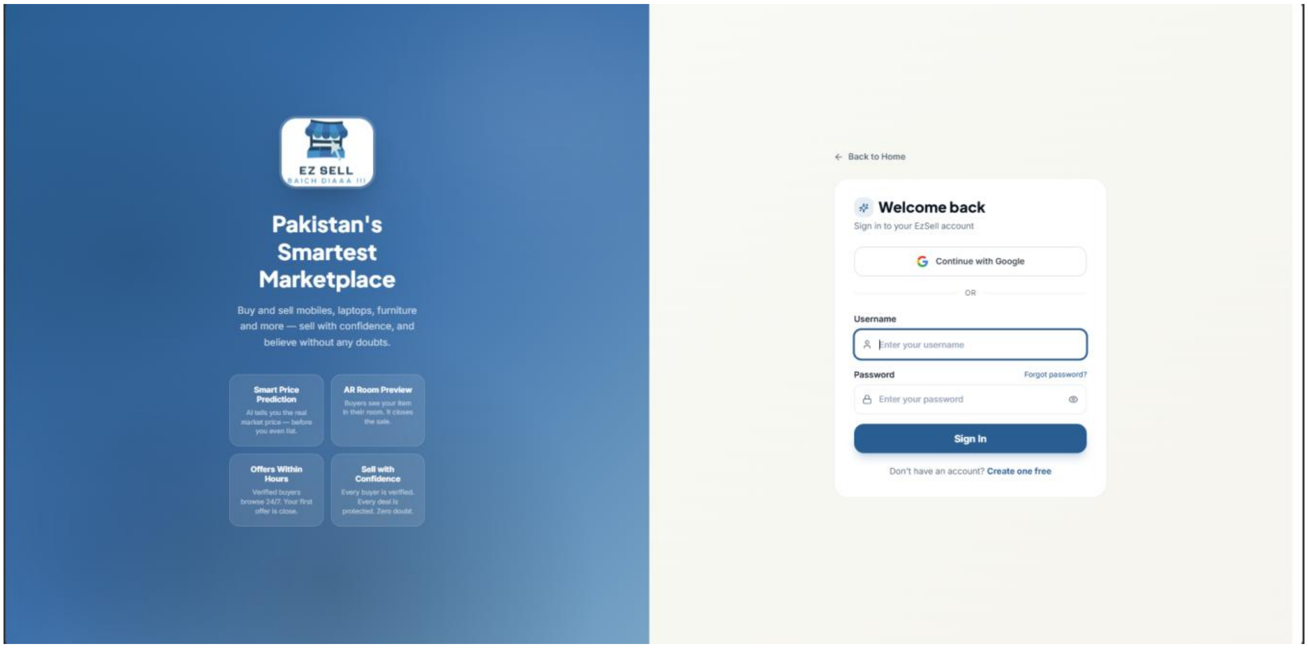
The EzSell platform utilizes an Object-Oriented (OO) data model mapped via SQLAlchemy ORM. The following table alphabetically lists the primary core entities, their attributes, methods, and descriptive purpose.

Table 15: Object-Oriented Data Dictionary

Object / Entity	Attributes (Name : Type)	Methods (Name(Params))	Description
Conversation	id: UUID listing_id: UUID buyer_id: UUID	get_messages(limit: Int) mark_read(user_id: UUID)	Groups messages between a buyer and seller for a specific listing.
Favorite	id: Int user_id: UUID listing_id: UUID created_at: DateTime	None	Association object tracking users' bookmarked listings.
Listing	id: UUID title: String description: Text price: Float status: Enum predicted_price: Float category: Enum images: JSON specs: JSON	validate_ai_price() update_status(new_status: Enum) check_fraud_hash()	Core inventory item (Mobile, Laptop, Furniture) representing an ad.
Message	id: Int conv_id: UUID sender_id: UUID content: Text read: Boolean timestamp: DateTime	None	Individual plaintext chat messages transmitted via WebSockets.
User	id: UUID email: String password_hash: String phone: String role: Enum avatar_url: String is_active: Boolean	verify_otp(code: String) generate_jwt() -> String update_profile(data: Dict)	System actor representing a buyer, seller, or administrator.

4.3 Human Interface Design

4.3.1 Login/Signup Screen


EZ SELL
BAICH DIAAA !!!
Back to Home
Continue with Google
OR
Username
Enter your username
Password
Forgot password?
Enter your password
Sign In
Don't have an account? Create one free

Pakistan's Smartest Marketplace

Buy and sell mobiles, laptops, furniture and more — sell with confidence, and believe without any doubts.

Smart Price Prediction

AI tells you the real market price — before you even list.

AR Room Preview

Buyers see your item in their room. It closes the sale.

Offers Within Hours

Verified buyers browse 24/7. Your first offer is close.

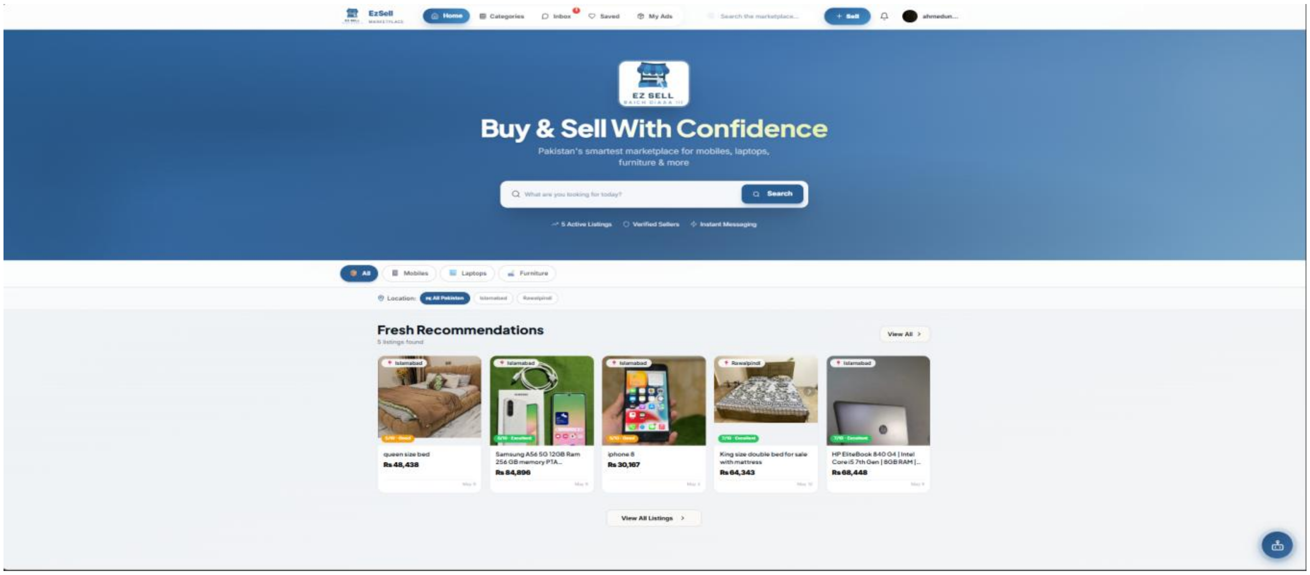
Sell with Confidence

Every buyer is verified. Every deal is protected. Zero doubt.

Welcome back

Sign in to your EzSell account

Figure 15 Login/Signup Page


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
EZ SELL
BAICH DIAAA !!!
What are you looking for today?
Search
5 Active Listings
Verified Sellers
Instant Messaging
All
Mobiles
Laptops
Furniture
Location:
All Pakistan
Islamabad
Rawalpindi
View All
Islamabad
Islamabad
Islamabad
Rawalpindi
Islamabad
View All Listings

Buy & Sell With Confidence

Pakistan's smartest marketplace for mobiles, laptops, furniture & more

Fresh Recommendations

5 listings found

queen size bed
Rs 48,438

Samsung A56 5G 12GB Ram 256 GB memory PTA...
Rs 84,896

iphone 8
Rs 30,167

King size double bed for sale with mattress
Rs 64,343

HP EliteBook 840 G4 | Intel Core i5 7th Gen | 8GB RAM |...
Rs 68,448

Figure 16 AD Listings Dashboard

4.3.2 AR View

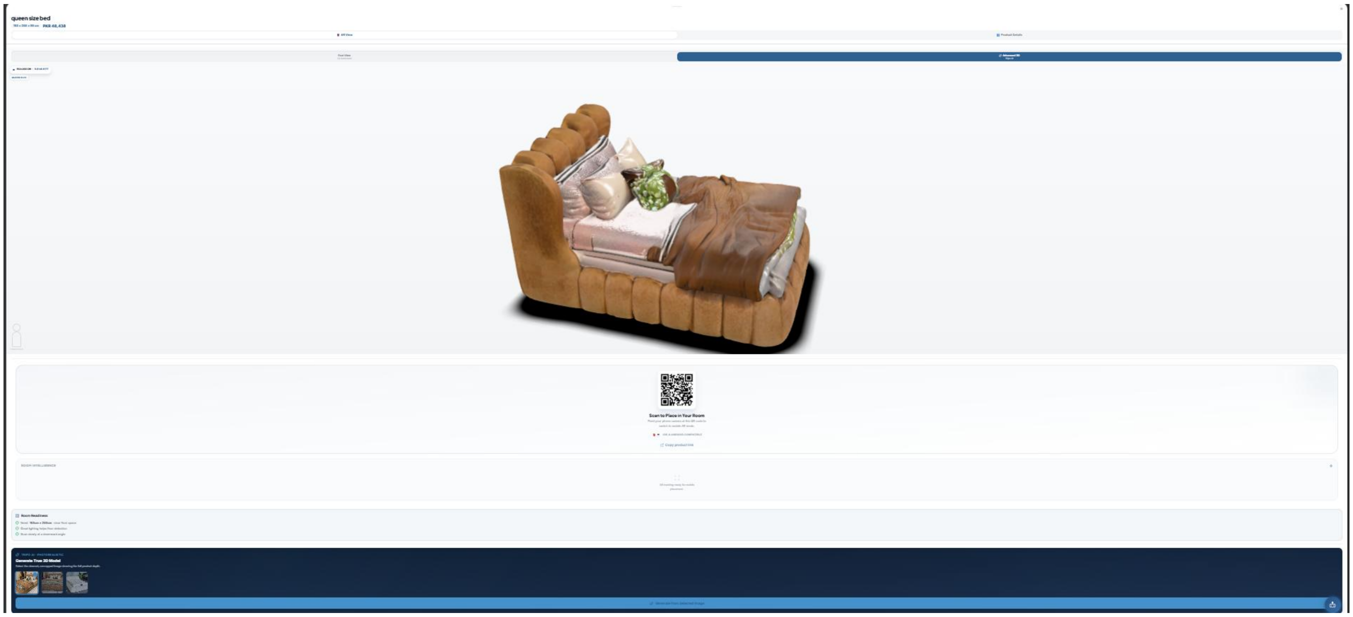
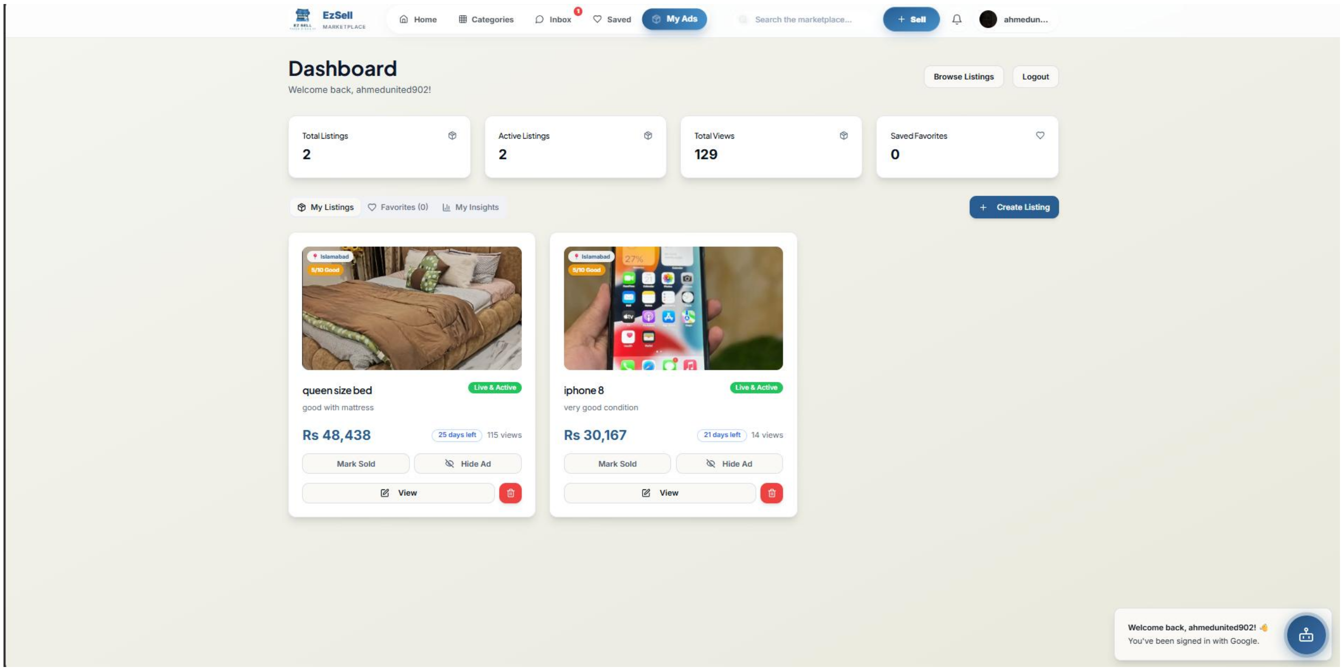

queen size bed
PKR 48,438
AR View
Scan to Place in Your Room
Copy product link

Figure 17 AR Furniture view

4.3.3 User Dashboard


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Browse Listings
Logout
My Listings
Favorites (0)
My Insights
+ Create Listing
Islamabad
5/10 Good
Live & Active
Mark Sold
Hide Ad
View
Islamabad
5/10 Good
Live & Active
Mark Sold
Hide Ad
View

Dashboard

Welcome back, ahmedunited902!

Total Listings	Active Listings	Total Views	Saved Favorites
2	2	129	0

queen size bed

good with mattress

Rs 48,438 25 days left 115 views

iphone 8

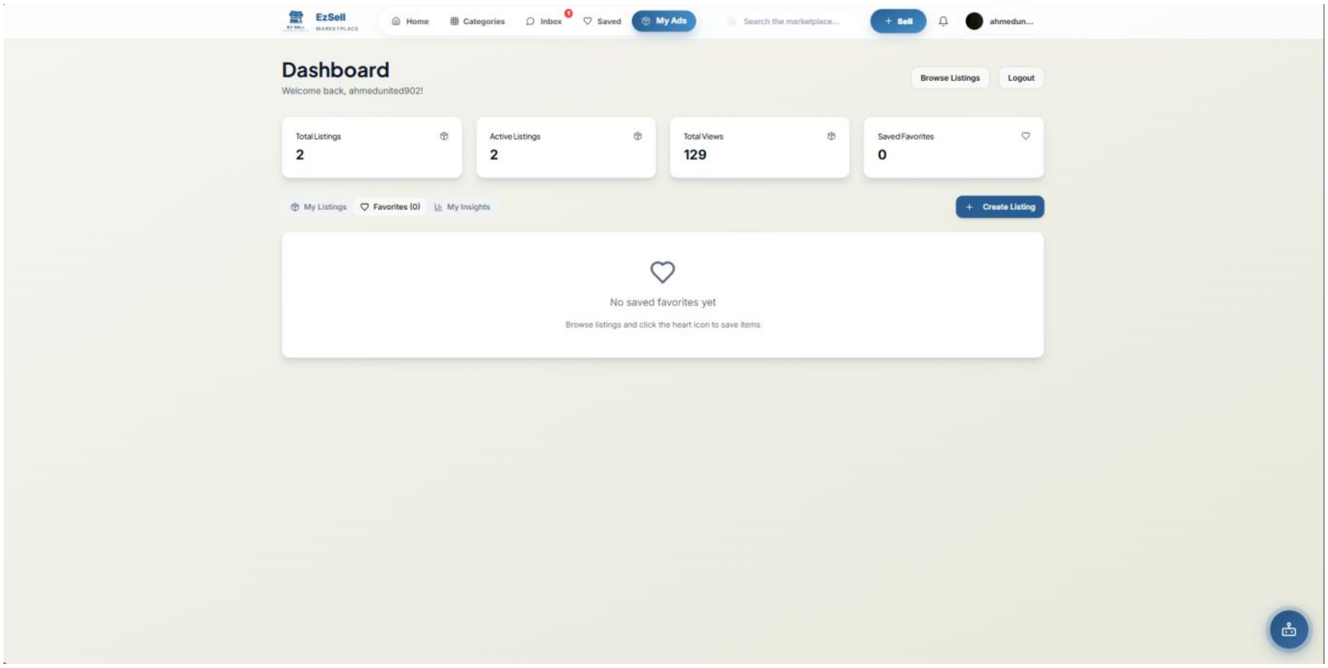
very good condition

Rs 30,167 21 days left 14 views

Welcome back, ahmedunited902! 👋

You've been signed in with Google.

Figure 18 User Dashboard


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Dashboard
Browse Listings
Logout
Total Listings
Active Listings
Total Views
Saved Favorites
My Listings
Favorites (0)
My Insights
+ Create Listing

Welcome back, ahmedunited902!

Total Listings: 2

Active Listings: 2

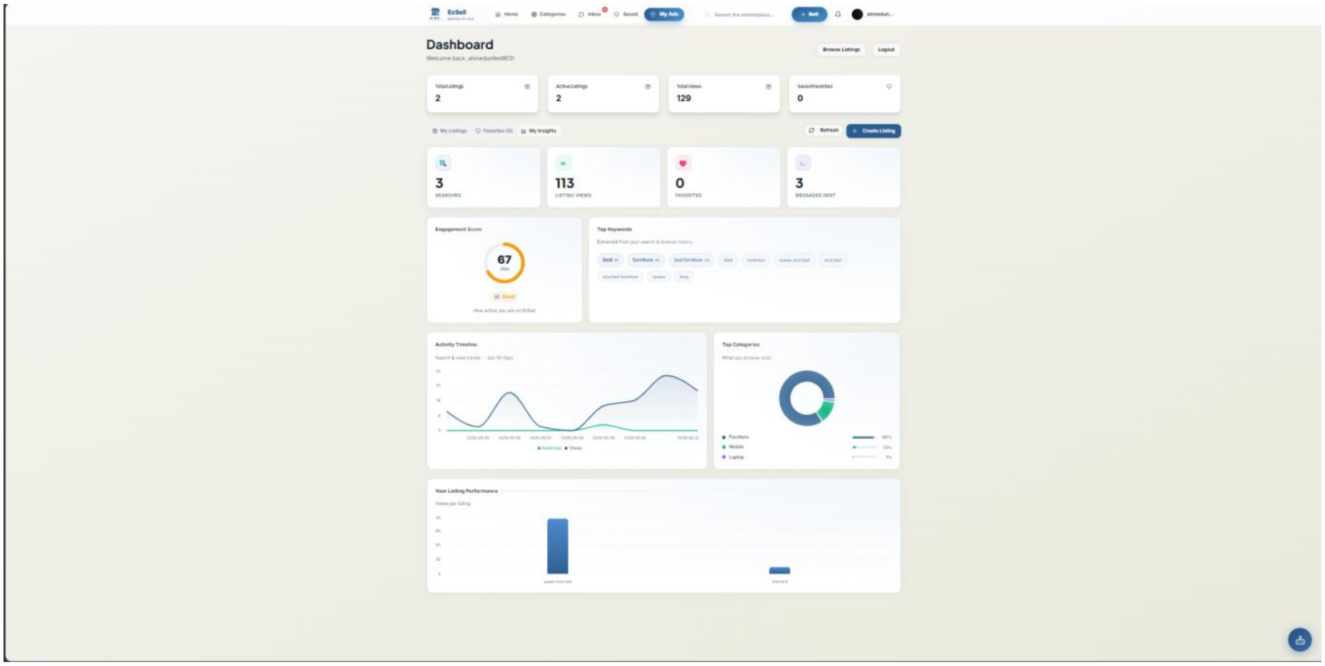
Total Views: 129

Saved Favorites: 0

No saved favorites yet

Browse listings and click the heart icon to save items

Figure 19 User Dashboard: Saved


EzSell
Home
Categories
Inbox
Saved
My Ads
+ Sell
Dashboard
Browse Listings
Logout
Total Listings
Active Listings
Total Views
Saved Favorites
My Listings
Favorites (0)
My Insights
Refresh
+ Create Listing
SEARCHES
LISTING VIEWS
FAVORITES
MESSAGES SENT
Engagement Score
67
Good
Top Keywords
bed
furniture
bed furniture
Activity Timeline
Searches
Views
Top Categories
Furniture
Mobile
Laptop
Your Listing Performance

Total Listings: 2

Active Listings: 2

Total Views: 129

Saved Favorites: 0

3 SEARCHES

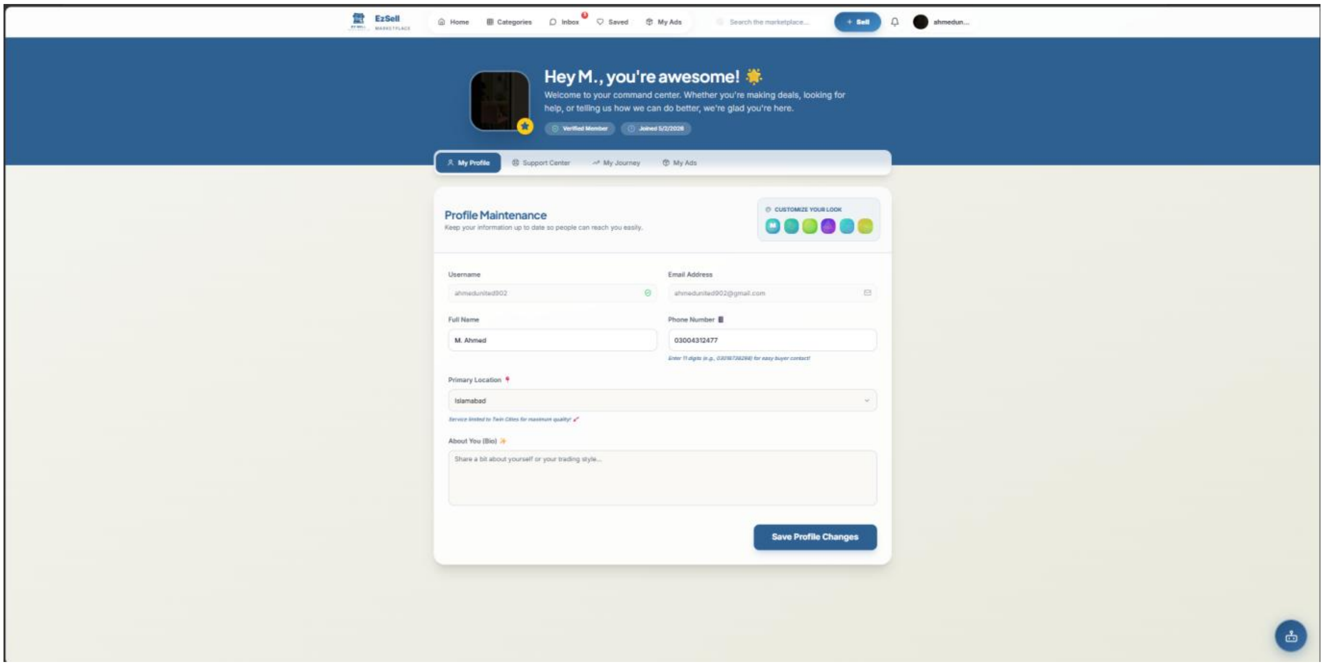
113 LISTING VIEWS

0 FAVORITES

3 MESSAGES SENT

Figure 20 User Dashboard: User Insights

4.3.4 User Profile


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Verified Member
Joined 5/2/2026
My Profile
Support Center
My Journey
My Ads
CUSTOMIZE YOUR LOOK
Save Profile Changes

Hey M., you're awesome!

Welcome to your command center. Whether you're making deals, looking for help, or telling us how we can do better, we're glad you're here.

Profile Maintenance

Keep your information up to date so people can reach you easily.

Username: ahmedunited902

Email Address: ahmedunited902@gmail.com

Full Name: M. Ahmed

Phone Number: 03004312477

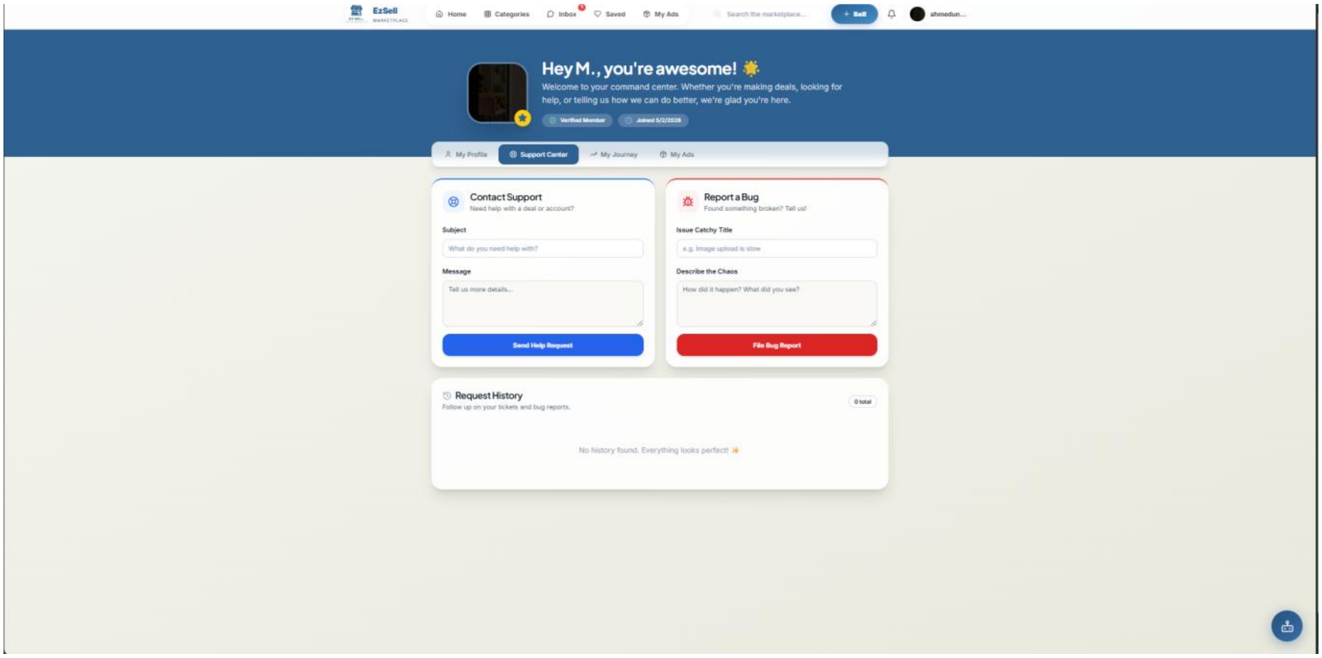
Enter 11 digits (e.g., [illegible]) for easy buyer contact!

Primary Location: Islamabad

Service limited to Twin Cities for maximum quality!

About You (Bio): Share a bit about yourself or your trading style...

Figure 21 User Profile


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Verified Member
Joined 5/2/2026
My Profile
Support Center
My Journey
My Ads
Send Help Request
File Bug Report
0 total

Hey M., you're awesome!

Welcome to your command center. Whether you're making deals, looking for help, or telling us how we can do better, we're glad you're here.

Contact Support

Need help with a deal or account?

Subject: What do you need help with?

Message: Tell us more details...

Report a Bug

Found something broken? Tell us!

Issue Catchy Title: e.g. Image upload is slow

Describe the Chaos: How did it happen? What did you see?

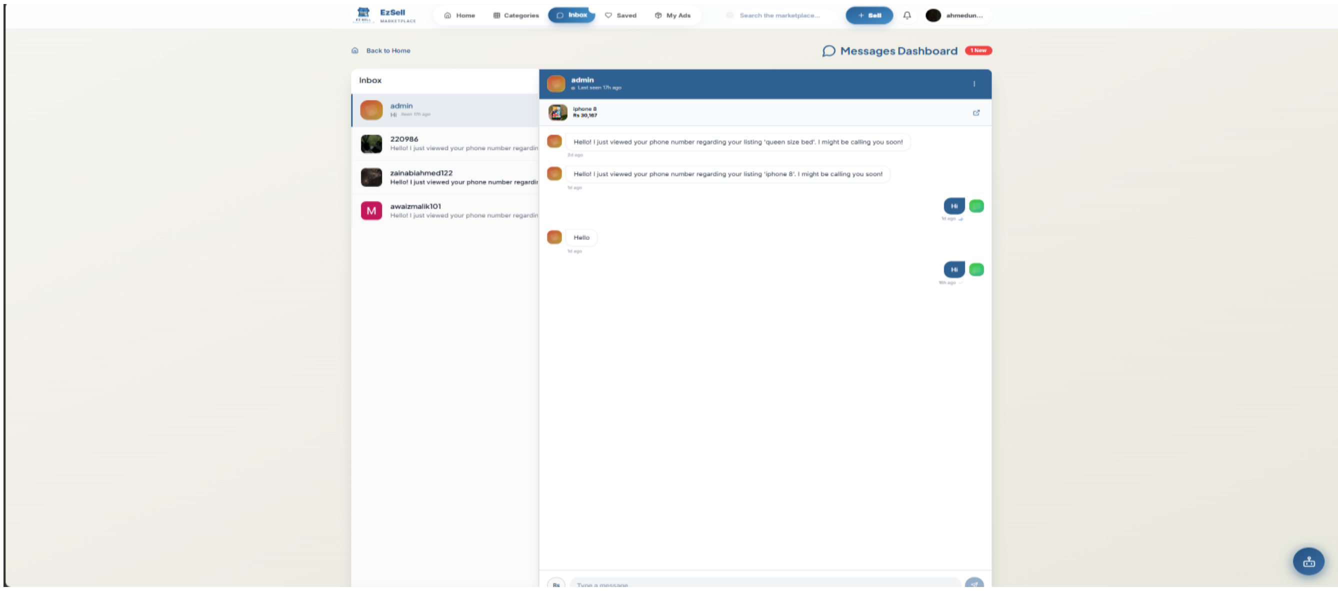
Request History

Follow up on your tickets and bug reports.

No history found. Everything looks perfect!

Figure 22 Bug Report page

4.3.5 In-App Chat Interface


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Back to Home
Messages Dashboard
1 New
Inbox
Last seen 17h ago
Type a message

admin
Hi

220986
Hello! I just viewed your phone number regardin

zainabiahmed122
Hello! I just viewed your phone number regardir

awaizmalik101
Hello! I just viewed your phone number regardin

Iphone 8
Rs 30,167

Hello! I just viewed your phone number regarding your listing 'queen size bed'. I might be calling you soon!

Hello! I just viewed your phone number regarding your listing 'iphone 8'. I might be calling you soon!

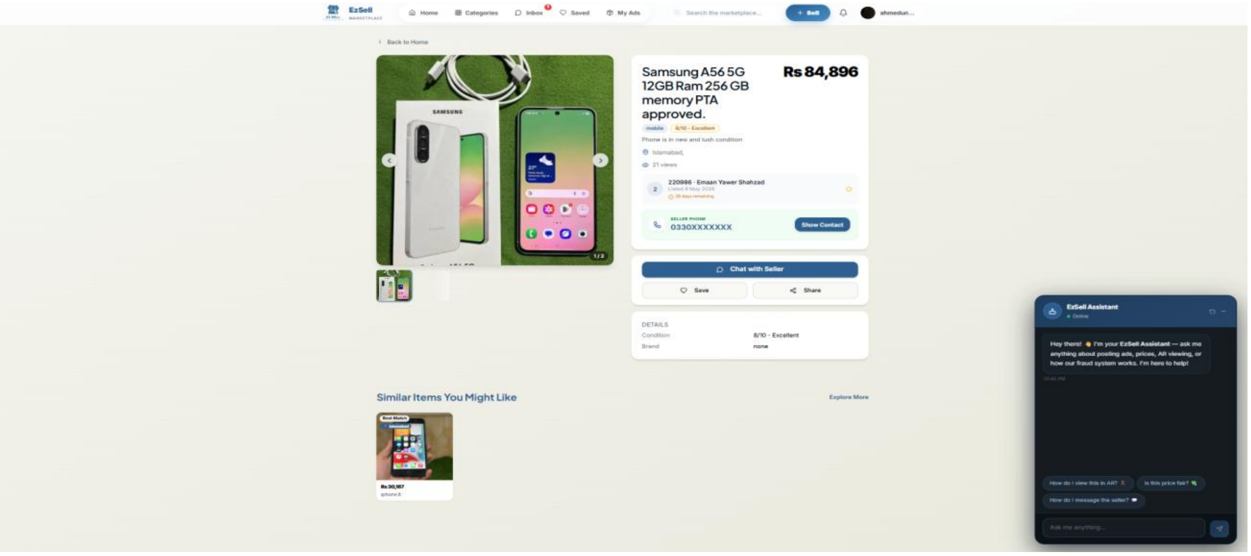
Hi

Hello

Hi

Figure 23 In App Chat Interface

4.3.6 Add Posted Interface and Built-in chatbot


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
My Ads
Search the marketplace...
+ Sell
ahmedun...
Back to Home
SAMSUNG
1/2
Show Contact
Chat with Seller
Save
Share
Similar Items You Might Like
Explore More
Rs 30,167
iphone 8
EzSell Assistant
Online
Ask me anything...

Samsung A56 5G 12GB Ram 256 GB memory PTA approved.

Rs 84,896

mobile
8/10 - Excellent

Phone is in new and lush condition

Islamabad,

21 views

220986 · Emaan Yawer Shahzad

SELLER PHONE
0330XXXXXXX

DETAILS

Condition	8/10 - Excellent
Brand	none

Hey there! 👋 I'm your EzSell Assistant — ask me anything about posting ads, prices, AR viewing, or how our fraud system works. I'm here to help!

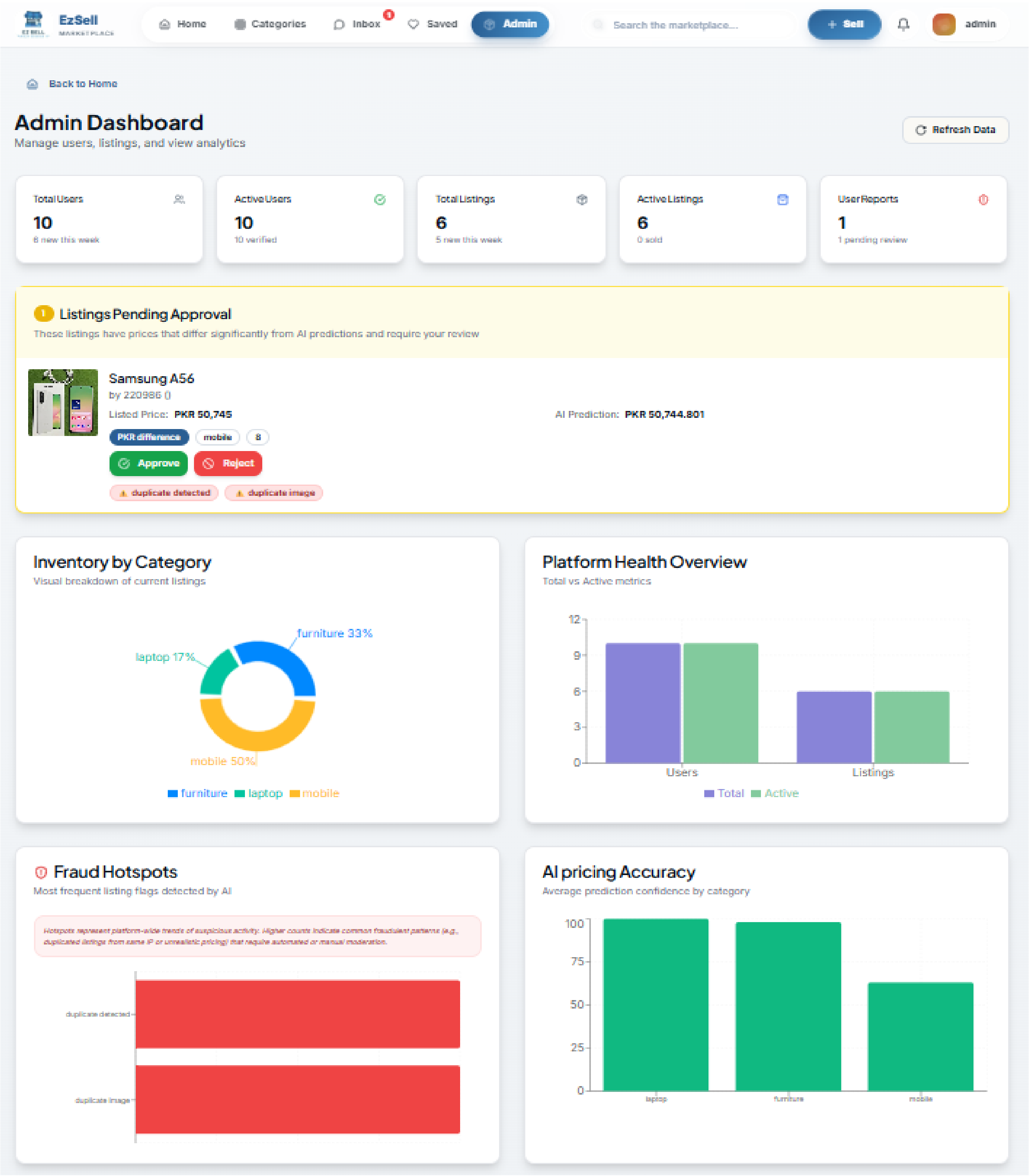
How do I view this in AR?

Is this price fair?

How do I message the seller?

Figure 24 Add Posted Interface and Built in chatbot

4.3.7 Admin Dashboard


EzSell
MARKETPLACE
Home
Categories
Inbox
Saved
Admin
Search the marketplace...
+ Sell
admin
Back to Home
Refresh Data
PKR difference
mobile
8
Approve
Reject
furniture 33%
laptop 17%
mobile 50%
furniture
laptop
mobile
12
9
6
3
0
Users
Listings
Total
Active
duplicate detected
duplicate image
100
75
50
25
0
laptop
furniture
mobile

Admin Dashboard

Manage users, listings, and view analytics

Total Users 10 — 8 new this week

Active Users 10 — 10 verified

Total Listings 6 — 5 new this week

Active Listings 6 — 0 sold

User Reports 1 — 1 pending review

Listings Pending Approval

These listings have prices that differ significantly from AI predictions and require your review

Samsung A56

by 220986 ()

Listed Price: **PKR 50,745**

AI Prediction: **PKR 50,744.801**

duplicate detected

duplicate image

Inventory by Category

Visual breakdown of current listings

Platform Health Overview

Total vs Active metrics

Fraud Hotspots

Most frequent listing flags detected by AI

Hotspots represent platform-wide trends of suspicious activity. Higher counts indicate common fraudulent patterns (e.g., duplicated listings from same IP or unrealistic pricing) that require automated or manual moderation.

AI pricing Accuracy

Average prediction confidence by category

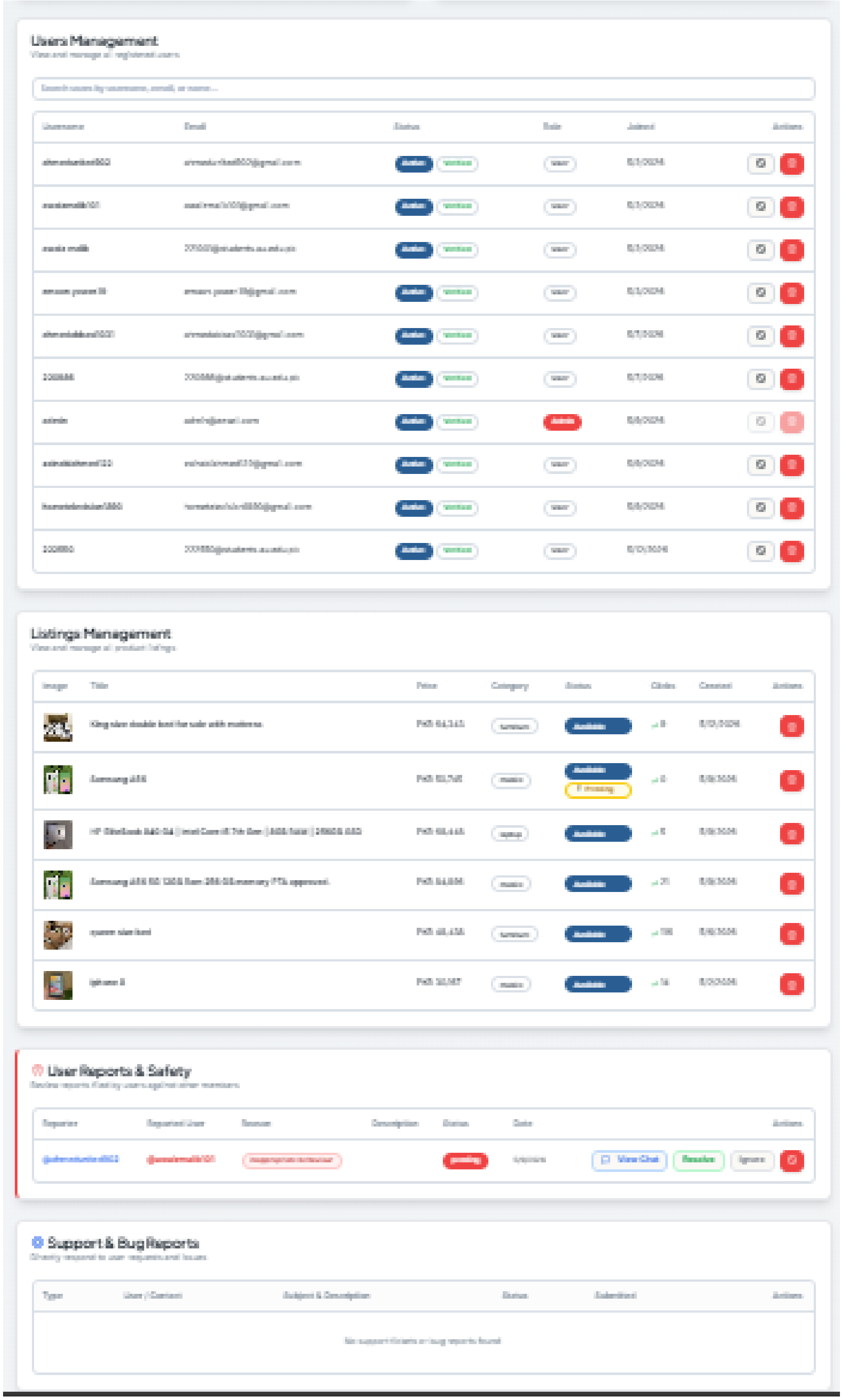
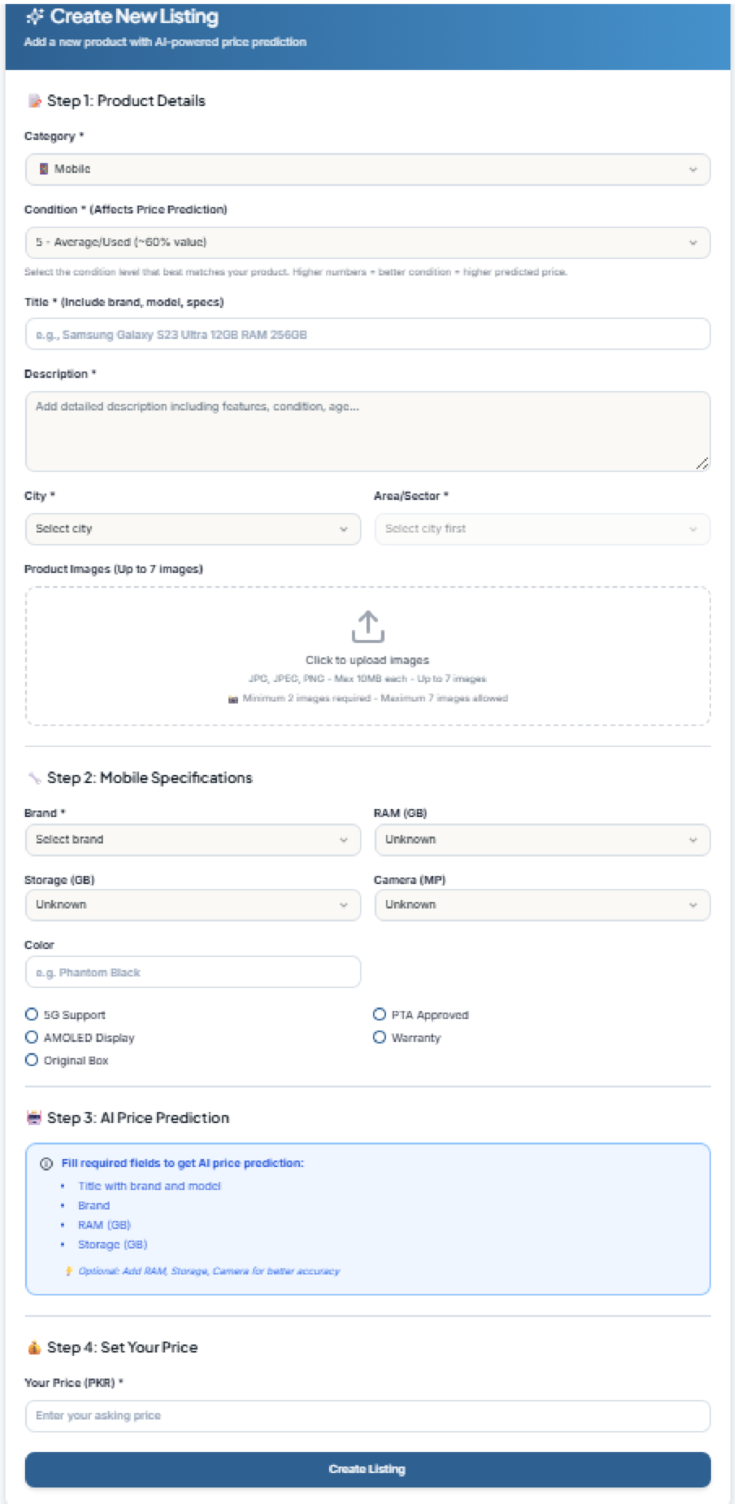

Users Management
Listings Management
User Reports & Safety
Support & Bug Reports

Figure 25 Admin Dashboard

4.3.8 Create New Listing



Create New Listing

Add a new product with AI-powered price prediction

Step 1: Product Details

Category *: Mobile

Condition * (Affects Price Prediction): 5 - Average/Used (~60% value)

Select the condition level that best matches your product. Higher numbers = better condition = higher predicted price.

Title * (Include brand, model, specs): e.g., Samsung Galaxy S23 Ultra 12GB RAM 256GB

Description *: Add detailed description including features, condition, age...

City *: Select city

Area/Sector *: Select city first

Product Images (Up to 7 images)

Click to upload images

JPG, JPEG, PNG - Max 10MB each - Up to 7 images

Minimum 2 images required - Maximum 7 images allowed

Step 2: Mobile Specifications

Brand *: Select brand

RAM (GB): Unknown

Storage (GB): Unknown

Camera (MP): Unknown

Color: e.g. Phantom Black

○ 5G Support
○ AMOLED Display
○ Original Box
○ PTA Approved
○ Warranty

Step 3: AI Price Prediction

Fill required fields to get AI price prediction:

- Title with brand and model
- Brand
- RAM (GB)
- Storage (GB)

Optional: Add RAM, Storage, Camera for better accuracy

Step 4: Set Your Price

Your Price (PKR) *: Enter your asking price

Figure 26 Create New Listing Page

5 Implementation

5.1 Algorithm

This section outlines the detailed pseudocode logic driving the major intelligent modules of the system. The algorithms are represented with standard programmatic semantics (e.g., assignment, loops, conditionals) to accurately reflect their operation.

5.1.1 Module 1: User Authentication & Account Management

Table 21: Algorithm 1 - User Authentication (JWT Generation)

Algorithm 1: JWT_Generation

Input: User Credentials (email, password)

Output: JWT Token (string) or Error

```
user_record ← DB_Find_User(email)
if user_record == null then
    return Error('Invalid credentials')
end if
is_valid ← bcrypt.verify(password, user_record.password_hash)
if is_valid == false then
    return Error('Invalid credentials')
end if
payload ← { 'sub': user_record.id, 'exp': current_time() + 24_hours }
token ← HS256_Sign(payload, SECRET_KEY)
return token
```

5.1.2 Module 2: Product Listings & Ad Management

Table 22: Algorithm 2 - Listing Category Inference

Algorithm 2: Metadata_Inference

Input: Title (string), Description (string)

Output: Extracted Metadata (Dictionary)

```
Metadata ← {}
title_tokens ← split(lower(Title), ' ')
foreach token in title_tokens do:
    if token matches Regex('[0-9]+GB') then
        Metadata['RAM'] ← token
    end if
    if token matches Regex('(iphone|samsung|dell|hp)') then
        Metadata['Brand'] ← extract_brand(token)
    end if
end foreach
return Metadata
```

5.1.3 Module 3: AI Price Prediction Pipeline

Table 23: Algorithm 3 - AI Price Prediction

Algorithm 3: AI_Price_Prediction_Pipeline

Input: Listing object L with attributes (title, specs, category)

Output: Final Predicted Price (Final_Price), Confidence Score (Conf)

```
weights ← {XGB: 0.35, LGBM: 0.35, RF: 0.15, GBC: 0.15}
ML_Base ← (weights.XGB * Predict_XGB(L)) + (weights.LGBM * Predict_LGBM(L)) + (weights.RF * Predict_RF(L)) + (weights.GBC * Predict_GBC(L))
query_string ← concatenate('site:olx.com.pk ', L.category, ' ', L.title)
Raw_URLs ← DuckDuckGo_Search(query_string, max_results=10)
Scraped_Prices ← []
foreach url in Raw_URLs do:
    price ← Groq_LLM_Extract_Price(Scrape_HTML(url))
    if price is valid then
        Scraped_Prices.append(price)
    end if
end foreach
if len(Scraped_Prices) < 3 then
    Final_Price ← ML_Base
    Conf ← 0.60
    return Final_Price, Conf
end if
Q1 ← Calculate_Percentile(Scraped_Prices, 25)
Q3 ← Calculate_Percentile(Scraped_Prices, 75)
IQR ← Q3 - Q1
Lower_Bound ← Q1 - (1.5 * IQR)
Upper_Bound ← Q3 + (1.5 * IQR)
Filtered_Prices ← []
foreach p in Scraped_Prices do:
    if (p ≥ Lower_Bound and p ≤ Upper_Bound) then
        Filtered_Prices.append(p)
    end if
end foreach
LLM_Market ← Average(Filtered_Prices)
Final_Price ← (ML_Base * 0.85) + (LLM_Market * 0.15)
Conf ← Calculate_Confidence(Variance(Filtered_Prices), ML_Base, LLM_Market)
return Final_Price, Conf
```

5.1.4 Module 4: Semantic Recommendations

Table 24: Algorithm 4 - Semantic Listing Ranking

Algorithm 4: Semantic_Ranking

Input: User interest vector U, Set of all active listings L

Output: Ranked list of recommended listings R

```
R ← []
threshold ← 0.5
foreach item in L do:
    if item.category == U.primary_category then
        item_vector ← SentenceTransformer_Encode(item.title + ' ' + item.description)
        similarity_score ← 1 - spatial.distance.cosine(U.vector, item_vector)
        if similarity_score ≥ threshold then
            R.append({listing: item, score: similarity_score})
        end if
    end if
end foreach
R ← Sort_Descending_By_Score(R)
return R.limit(20)
```

5.1.5 Module 5: Analytical Dashboard

Table 25: Algorithm 5 - Metrics Aggregation

Algorithm 5: Metrics_Aggregation

Input: Date Range (start_date, end_date)

Output: Aggregated Statistics Object

```
Stats ← {}
Stats.Total_Users ← DB_Count('Users', where='created_at BETWEEN start_date AND end_date')
Stats.Active_Listings ← DB_Count('Listings', where='status == Active')
Stats.Pending_Tickets ← DB_Count('Tickets', where='status == Open')
Stats.Daily_Signups ← DB_Group_By_Day('Users', start_date, end_date)
return Stats
```

5.1.6 Module 6: AR Customization Try-On

Table 26: Algorithm 6 - 3D GLB Model Generation

Algorithm 6: AR_3D_Model_Generation

Input: List of 2D images I, Category C

Output: GLB Model URL or Error

```
if C != 'Furniture' then
    return Error('AR Try-On restricted to Furniture category.')
end if
Primary_Image ← Select_Highest_Resolution(I)
Processed_Image ← Remove_Background_And_Center(Primary_Image)
Task_ID ← Tripo_API_Submit(Processed_Image)
status ← 'Processing'
timeout ← 60 seconds
while (status == 'Processing' and timeout > 0) do:
    Wait(2 seconds)
    timeout ← timeout - 2
    status, Model_URL ← Tripo_API_Check_Status(Task_ID)
    if status == 'Success' then
        return Model_URL
    else if status == 'Failed' then
        return Error('3D Generation Failed.')
    end if
end while
return Error('Timeout exceeded.')
```

5.1.7 Module 7: Fraud Prevention Pipeline

Table 27: Algorithm 7 - Fraud Detection and Blocking

Algorithm 7: Fraud_Prevention_Pipeline

Input: Listing L, Array of Images Imgs

Output: Boolean status (True = Approved, False = Rejected/Flagged)

```
Listing_Hash ← SHA256(L.title + L.price + L.category)
if (Database_Contains(Listing_Hash) == true) then
    Mark_As_Rejected(L, 'Duplicate Listing')
    return False
end if
foreach img in Imgs do:
    pred_class ← CLIP_Zero_Shot_Predict(img, ['mobile', 'laptop', 'furniture', 'random'])
    if (pred_class != L.category) then
        Mark_As_Rejected(L, 'Image Category Mismatch')
        return False
    end if
    img_hash ← calculate_dHash(img)
    matches ← Find_Similar_Hashes(img_hash, threshold=5)
    if (len(matches) > 0) then
        Mark_As_Rejected(L, 'Stolen/Reused Image')
        return False
    end if
end foreach
Scam_Keywords ← ['send money', 'western union', 'advance payment']
foreach word in Scam_Keywords do:
    if Regex_Match(L.description, word) then
        Mark_As_Flagged(L, 'Manual Review Required')
        return False
    end if
end foreach
Mark_As_Approved(L)
return True
```

5.1.8 Module 8: Support & Notifications

Table 28: Algorithm 8 - Support Ticket Creation

Algorithm 8: Ticket_Creation

Input: User U, Subject (string), Description (string)

Output: Ticket ID

```
if len(Description) < 10 then
    return Error('Description too short')
end if
Ticket ← Create_DB_Record('Tickets', user_id=U.id, subject=Subject, desc=Description, status='Open')
trigger_event('ticket_created', Ticket)
return Ticket.id
```

5.1.9 Module 9: Messaging & Inbox

Table 29: Algorithm 9 - WebSocket Message Routing

Algorithm 9: WebSocket_Routing

Input: Sender_ID, Receiver_ID, Message_Content, Listing_ID

Output: Boolean delivery status

```
Conv_ID ← get_or_create_conversation(Sender_ID, Receiver_ID, Listing_ID)
Msg ← store_message_in_db(Conv_ID, Sender_ID, Message_Content)
if is_user_connected(Receiver_ID) then
    socket ← get_active_websocket(Receiver_ID)
    send_payload(socket, Msg)
    Msg.delivered ← true
else
    increment_unread_badge(Receiver_ID)
end if
return Msg.delivered
```

5.1.10 Module 10: Favorites & User Dashboard

Table 30: Algorithm 10 - Favorites Toggle Logic

Algorithm 10: Favorites_Toggle

Input: User_ID, Listing_ID

Output: New State ("Added" or "Removed")

```
fav_record ← DB_Query('Favorites', user_id=User_ID, listing_id=Listing_ID)
if fav_record exists then
    DB_Delete(fav_record)
    return 'Removed'
else
    DB_Insert('Favorites', user_id=User_ID, listing_id=Listing_ID)
    return 'Added'
end if
```

5.1.11 Module 11: AI Chatbot Assistant

Table 31: Algorithm 11 - Context Injection

Algorithm 11: AI_Context_Injection

Input: User Query Q

Output: Streamed Response Generator

```
market_data ← Load_CSV('olx_market_summary.csv')
system_prompt ← 'You are EzSell Assistant. Use this data: ' + format_to_string(market_data)
full_prompt ← system_prompt + '\nUser: ' + Q
stream ← Groq_API_Call('llama-3.1-8b-instant', full_prompt, stream=True)
foreach token in stream do:
    yield token
end foreach
```

5.2 External APIs/SDKs

Table 32 Details of APIs used in the project

Name of API and version	Description of API	Purpose of usage	List down the API endpoint/function/class in which it is used
Groq API (v1)	Ultra-fast LLM inference API	llama-3.3-70b for price validation; llama-3.1-8b for chatbot streaming.	https://api.groq.com/openai/v1/chat/completions
Tripo AI (Version V2)	Image-to-3D generation service	Generates GLB 3D models from furniture photos for AR try-on.	https://api.tripo3d.ai/v2/openapi/task
Cloudinary API (v1_1)	Cloud image and video management solution	Uploading Product Images on Cloudinary server	https://api.cloudinary.com/v1_1/demo/image/upload
DuckDuckGo DDGS (Latest)	Python search scraping library	Scrapes live OLX Pakistan links to retrieve recent market prices.	duckduckgo_search.DDGS.text
Google OAuth 2.0 (v2)	Authentication and authorization framework	Enables users to securely log in using their Google accounts.	https://oauth2.googleapis.com/token

5.3 User Interface

The React frontend uses Shadcn UI and Tailwind CSS. The Home Feed implements infinite scrolling. The AR Viewer uses Google model-viewer for native spatial visualization.

5.4 Deployment

During development, we ran and tested the whole system on our local machines. The project had two main parts: the backend API and the frontend web application. Both were connected and tested together in a local setup.

5.4.1 Backend Environment

- Framework: FastAPI
- Runtime: Python 3.13
- Server Used: Uvicorn (development server)
- Hosting Mode: Localhost (http://127.0.0.1:8000/)
- Main Dependencies: fastapi, uvicorn, sqlalchemy, pydantic, httpx, python-jose, passlib[bcrypt], google-genai, python-multipart, python-dotenv, pandas, numpy, matplotlib, prophet, transformers, torch, groq, bcrypt

5.4.2 Frontend Environment

- Framework: React.js
- Runtime Environment: Node.js (LTS version)
- Local Hosting URL: http://localhost:8080/
- Package Manager: npm
- Primary Dependencies: React Router, Axios, Redux / Context API, Tailwind CSS, Utility Libraries

5.4.3 Testing Setup

- Simultaneous Local Execution: Both the FastAPI backend and the React.js frontend were run together on the local development system.
- API Communication: The frontend interacted with the backend using REST API calls over local URLs.
- Modules Tested in Local Environment: User Authentication, Product Listings, AI Price Prediction, Dashboards, 3D AR Module, Database CRUD Operations.

For production, the EzSell platform is deployed on Amazon Web Services (AWS) to guarantee high availability, low-latency access, and robust computational performance.

5.4.4 Cloud Infrastructure Details

Specification	Detail
Hosting Service	Amazon Web Services (AWS) Elastic Compute Cloud (EC2)
Instance ID	i-037ea3edb54596c83
Instance Name	ezsell backend
Region & Availability Zone	eu-north-1 (Europe - Stockholm) / eu-north-1c
Status	Running

5.4.5 Compute and Storage Specifications

Component	Specification Details
Instance Type	c7i-flex.large (x86_64 Architecture)
CPU	1 core with 2 threads per core
Operating System	Ubuntu 26.04 LTS (ubuntu-resolute-26.04-amd64-server-20260421)
Storage Volume	16 GiB gp3 SSD (Root Volume)
Storage Performance	3,000 IOPS, 125 MB/s Throughput
Encryption	Not encrypted, Delete on Termination Enabled

5.4.6 Network and Security Configuration

Parameter	Configuration Details
VPC and IPs	Default VPC (vpc-092755086093495e8) | Private IP: 172.31.0.165 | Public IP: 13.51.45.172
Public DNS	ec2-13-51-45-172.eu-north-1.compute.amazonaws.com
Security Group	launch-wizard-1
Inbound Rules	Port 22 (SSH), Port 80 (HTTP), Port 443 (HTTPS) - Open to all IPs (0.0.0.0/0)
Outbound Rules	All outbound traffic allowed

5.4.7 Access Information and Software Stack

- SSH Access Command: ssh -i ezsell.pem ubuntu@13.51.45.172
- Web Server Access: http://13.51.45.172 or https://13.51.45.172
- Web Server Software: Uvicorn ASGI Server managed by Gunicorn process workers.

6 Testing and Evaluation

6.1 API & Module Endpoint Testing

The following tables detail the endpoint-level test cases and their execution outcomes across the core modules of the application.

6.1.1 M1 — Profile Management Tests

Table 33: M1 Profile Management Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M1-1	Register new user with email & phone	POST /auth/register	201 Created, user in DB	Pass
T-M1-2	Login with correct credentials	POST /auth/login	200 OK, JWT returned	Pass
T-M1-3	Login with wrong password	POST /auth/login	401 Unauthorized	Pass
T-M1-4	Google OAuth sign-in flow	Browser OAuth redirect	Token set, user created/fetched	Pass
T-M1-5	Access secured endpoint without token	GET /users/me	401 Unauthorized	Pass
T-M1-6	Update profile fields	PATCH /users/me	200 OK, fields updated	Pass
T-M1-7	Upload avatar image	POST /users/me/avatar	200 OK, avatar_url set	Pass
T-M1-8	Admin flag verification	GET /users/me as admin	is_admin: true in response	Pass

6.1.2 M2 — Product Listings Tests

Table 34: M2 Product Listings Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M2-1	Create listing with 3 images	POST /listings/	201 Created	Pass
T-M2-2	Fetch listing by ID	GET /listings/{id}	200 OK, full listing data	Pass
T-M2-3	Edit listing title	PATCH /listings/{id}	200 OK, title updated	Pass
T-M2-4	Delete listing as owner	DELETE /listings/{id}	200 OK, removed from DB	Pass
T-M2-5	Hide listing (toggle)	PATCH /listings/{id}/toggle-visibility	is_active: false	Pass
T-M2-6	Mark listing as sold	PATCH /listings/{id}/mark-sold	is_sold: true	Pass
T-M2-7	Submit listing for approval	POST /listings/	approval_status: pending	Pass
T-M2-8	Admin approve listing	PATCH /approvals/{id}/approve	approval_status: approved	Pass
T-M2-9	Search by category	GET /listings/?category=mobile	Filtered results returned	Pass
T-M2-10	AI title validation	POST /listings/validate-title	Valid/invalid flag returned	Pass

6.1.3 M3 — Price Prediction Tests

Table 35: M3 Price Prediction Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M3-1	Predict price for mobile phone (ML)	POST /predictions/mobile	Price range + confidence score	Pass
T-M3-2	Predict price for laptop (ML)	POST /predictions/laptop	Price range + confidence score	Pass
T-M3-3	Predict price for furniture (ML)	POST /predictions/furniture	Price range + confidence score	Pass
T-M3-4	Groq LLM price estimation	POST /predictions/advanced	LLM-sourced price + OLX snippets	Pass
T-M3-5	OLX DuckDuckGo scraping	Internal service call	Prices extracted and IQR-filtered	Pass
T-M3-6	Prediction on create form	Frontend UX	Suggested price shown to seller	Pass
T-M3-7	Confidence score < 50% flagged	Frontend UX	Low confidence warning shown	Pass

6.1.4 M4 — Recommendations Tests

Table 36: M4 Recommendations Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M4-1	Log view activity	POST /recommendations/activity	200 OK, activity recorded	Pass
T-M4-2	Get personalized feed	GET /recommendations/	Returns interest-matched listings	Pass
T-M4-3	Get trending for anonymous	GET /recommendations/trending	Returns top-viewed listings	Pass
T-M4-4	Filter by price range	GET /listings/?min_price=X&max_price=Y	Listings within range returned	Pass
T-M4-5	Semantic similarity search	GET /listings/?q=leather+sofa	NLP-ranked results returned	Pass

6.1.5 M5 — Analytical Dashboard Tests

Table 37: M5 Analytical Dashboard Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M5-1	Admin fetch dashboard stats	GET /analytics/overview	User/listing/revenue totals	Pass
T-M5-2	Admin view all users	GET /users/ (admin)	Full user list with details	Pass
T-M5-3	Admin view all listings	GET /listings/ (admin)	All listings regardless of status	Pass
T-M5-4	Admin reject listing with reason	PATCH /approvals/{id}/reject	Status set, rejection reason saved	Pass
T-M5-5	Admin view support tickets	GET /support/admin/tickets	All tickets with user info	Pass
T-M5-6	Support requests count card	Frontend UX	Correct total + open count shown	Pass

6.1.6 M6 — AR / Try-On Tests

Table 38: M6 AR / Try-On Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M6-1	Generate procedural 3D model	POST /products/{id}/assets/generate	GLB/USDZ URLs returned	Pass
T-M6-2	Launch WebAR viewer	Frontend UX (/ar/{id})	model-viewer loads GLB	Pass
T-M6-3	Trigger Tripo AI generation	POST /products/{id}/assets/generate-3d	Task ID returned, polling starts	Pass
T-M6-4	Poll AI generation status	GET /products/{id}/assets/generate-3d/{task_id}	status: done, URLs available	Pass
T-M6-5	Admin manual GLB upload	POST /products/{id}/assets/upload-glb	GLB URL stored in DB	Pass
T-M6-6	iOS USDZ AR QuickLook	iOS Safari	Native AR viewer launched	Pass
T-M6-7	Fixed scale lock	Frontend UX	Model does not resize freely	Pass

6.1.7 M7 — Fraud Prevention Tests

Table 39: M7 Fraud Prevention Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M7-1	Detect duplicate listing	POST duplicate listing	listing_hash collision detected	Pass
T-M7-2	Detect duplicate image	POST listing with reused image	image_hash collision flagged	Pass
T-M7-3	Fraud flags stored	DB inspection	JSON array in fraud_flags column	Pass
T-M7-4	Flagged listing surfaced to admin	Admin Dashboard	Flagged items appear in review queue	Pass
T-M7-5	Reject with documented reason	PATCH /approvals/{id}/reject + reason	rejection_reason field saved in DB	Pass

6.1.8 M8 — Support & Notifications Tests

Table 40: M8 Support & Notifications Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M8-1	Submit support ticket	POST /support/tickets	201 Created, ticket in DB	Pass
T-M8-2	Submit bug report	POST /support/tickets (type: bug)	201 Created	Pass
T-M8-3	Admin fetch all tickets	GET /support/admin/tickets	All tickets with user info returned	Pass
T-M8-4	Admin update ticket to working	PATCH /support/admin/tickets/{id}/status	Status updated, notification created	Pass
T-M8-5	Admin update ticket to done	PATCH /support/admin/tickets/{id}/status	Status updated, notification created	Pass
T-M8-6	User fetch notifications	GET /notifications	Notification list returned	Pass
T-M8-7	Unread count endpoint	GET /notifications/unread/count	Correct count returned	Pass
T-M8-8	Mark notification as read	POST /notifications/{id}/read	is_read: true set in DB	Pass
T-M8-9	Mark all notifications as read	POST /notifications/read-all	All user notifications marked read	Pass
T-M8-10	Bell badge shows unread count	Frontend UX	Badge visible with correct number	Pass
T-M8-11	Notification popover renders	Frontend UX	Alert list visible in dropdown	Pass
T-M8-12	Click notification navigates user	Frontend UX	User routed to /profile	Pass

6.1.9 M9 — Messaging Tests

Table 41: M9 Messaging Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M9-1	Send message to seller	POST /messages/	201 Created, stored in DB	Pass
T-M9-2	Fetch conversations list	GET /messages/conversations	All threads returned	Pass
T-M9-3	Read messages in a conversation	GET /messages/{conversation_id}	Message history returned	Pass
T-M9-4	Unread message count in nav	GET /messages/unread/count	Correct count in badge	Pass
T-M9-5	Mark conversation as read	POST /messages/{id}/read	is_read: true for all messages	Pass

6.1.10 M10 — Favorites & Dashboard Tests

Table 42: M10 Favorites & Dashboard Tests

Test ID	Test Case	Method / Layer	Expected Result	Outcome
T-M10-1	Add listing to favorites	POST /favorites/{id}	200 OK, saved in DB	Pass
T-M10-2	Remove listing from favorites	DELETE /favorites/{id}	200 OK, removed from DB	Pass
T-M10-3	Fetch all user favorites	GET /favorites/	Full favorites list returned	Pass
T-M10-4	Check if listing is favorited	GET /favorites/{id}/check	is_favorited: true/false	Pass
T-M10-5	Dashboard shows own listings	GET /listings/my	Only user's listings returned	Pass
T-M10-6	Dashboard status count cards	Frontend UX	Correct active/hidden/sold counts	Pass

6.2 Unit Testing

Unit testing checks individual components to ensure each one behaves as expected.

6.2.1 Unit Testing 1: Validate login functionality with correct and incorrect inputs.

Testing Objective: To ensure the login form is working correctly with valid and invalid inputs.

Table 43: Unit Testing 1 login Functionalities

No.	Test case	Attribute and value	Expected result	Result
1	Check valid login	username: ahmed Password: ahmed123	User is logged in and redirected to dashboard.	Pass
2	Invalid email format	username: ahme	Error, no such sort of username	Pass
3	Invalid password	Password: EZsell123	Dashboard unaccessible, login page refresh	Pass

6.2.2 Unit Testing 2: Listing Image Upload

Testing Objective: Ensure supported file types are accepted and invalid formats are rejected.

Table 44: Unit Testing 2 Listing Image Upload

No.	Test Case	Attribute and Value	Expected Result	Result
1	Upload supported image	image.jpg	File accepted and uploaded to storage	Pass
2	Upload unsupported format	file.txt	Wont show any other sort of file other than jpg, jpeg, png	Pass
3	Upload large file	big_image.webp (10MB)	Error	Pass

6.2.3 Unit Testing 3: Price Prediction

Testing Objective: Ensure that the AI price prediction module correctly processes features and returns a valid predicted price with confidence.

Table 45: Unit Testing 3 Price Prediction

No.	Test Case	Attribute and Value	Expected Result	Result
1	Predict price for mobile	title=Samsung s23, condition=used	Return predicted price + confidence	Pass
2	Predict price for laptop	brand=HP omen, core i3, RAM=8GB	Return predicted price + confidence	Pass
3	Missing required feature	condition = null	No Prediction	Pass

6.2.4 Unit Testing 4: Listing Creation (Post Ad)

Testing Objective: Ensure that new listings are created successfully with all required fields.

Table 46: Unit Testing 4 Listing Creation

No.	Test Case	Attribute and Value	Expected Result	Result
1	Valid listing	title, desc, price, category, image provided	Listing saved successfully	Pass
2	Missing title	title=null	Error	Pass
3	Price not numeric	price="ten thousand"	Error	Pass
4	Image missing	no images uploaded	Error	Pass

6.2.5 Unit Testing 5: User Dashboard Data Fetch

Testing Objective: Ensure that dashboard fetches correct analytics based on user_id.

Table 47: Unit Testing 5 Dashboard Data

No.	Test Case	Attribute and Value	Expected Result	Result
1	Fetch dashboard data	user_id valid	Returns totals (views, listings, contacts)	Pass
2	No listings	user with 0 listings	Show zero analytics	Pass
3	Invalid user_id	user_id = null	Error	Pass

6.2.6 Unit Testing 6: Admin Analytics Dashboard

Testing Objective: Verify correct aggregation of platform-wide metrics.

Table 48: Unit Testing 6 Admin Analytics

No.	Test Case	Attribute and Value	Expected Result	Result
1	Fetch system metrics	none	Return totals: users, listings, active listings	Pass
2	No users available	empty database	Display zero metrics, no errors	Pass

6.2.7 Unit Testing 7: AR Room Analysis Functions

Testing Objective: Ensure that AR room analysis helper functions correctly process images and detect room properties.

Table 49: Unit Testing 7 AR Analysis

No.	Test Case	Attribute and Value	Expected Result	Result
1	Detect dominant colors	room image with blue walls	Returns color: "Blue", hex: "#4A90E2"	Pass
2	Estimate room dimensions	room image with visible walls	Returns width, height, depth estimates	Pass
3	Detect room style	modern furniture in image	Returns style: "Modern", confidence: 85%	Pass
4	Calculate suitability	furniture vs room dimensions	Returns score: 75–95%	Pass
5	Invalid image format	file: not an image	Returns error: "Invalid image"	Pass

6.3 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the module is tested. This is to ensure that the system produced meets the specifications and requirements.

6.3.1 Functional Testing 1: Upload Listing Images

Testing Objective: Ensure that uploaded listing images/files are accepted for supported formats and rejected otherwise.

Table 50: Functional Testing 1

No.	Test case	Attribute and value	Expected result	Actual result	Result
1	Upload valid image	image.jpg	File accepted and uploaded to storage	Uploaded successfully	Pass
2	Upload unsupported file type	file.txt	Won’t allow	Didn’t allowed	Pass
3	Upload large file (>10MB)	Big image.webp	Display warning: "File size too large"	Warning shown	Pass

6.3.2 Functional Testing 2: AI Price Prediction

Testing Objective: Verify that the price prediction engine returns valid prices and confidence for different listings.

Table 51: Functional Testing 2

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	Predict price for mobile device	Category: Mobile, Brand: Samsung, RAM: 8GB, Storage: 128GB	Display predicted price along with confidence range	Prediction displayed correctly	Pass
2	Predict price for laptop	Category: Laptop, Brand: HP, Processor: Intel i5, RAM: 8GB	Display predicted price along with confidence range	Prediction displayed correctly	Pass
3	Prediction with high confidence	Well-defined and complete specifications	Display narrow confidence range	High confidence shown correctly	Pass
4	Auto-approval range display	Successful price prediction	Display green auto-approval price range	Auto-approval range displayed correctly	Pass
5	Re-prediction on field change	User modifies product specifications	System automatically triggers a new price prediction	Auto-prediction working correctly	Pass

6.3.3 Functional Testing 3: User & Admin Dashboard Analytics

Testing Objective: Ensure that dashboards correctly display metrics for users and admins.

Table 52: Functional Testing 3

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	User dashboard with listings	Valid user_id with existing listings	Display total listings, total views, and contact count	Metrics displayed correctly	Pass
2	User dashboard with no listings	User with zero listings	Display message: "No data available"	Message displayed correctly	Pass
3	Admin dashboard metrics	Complete system data available	Display total users, total listings, and active listings	Metrics displayed correctly	Pass
4	Admin dashboard with empty database	No users or listings in system	Display zero values for all metrics	Zero metrics displayed correctly	Pass
5	Admin dashboard top sellers	Users with multiple listings	Display top 5 sellers ranked by number of listings	Top sellers displayed correctly	Pass

6.3.4 Functional Testing 4: Listing Creation & Validation

Testing Objective: Ensure that listing creation, duplicate prevention, and price validation work correctly.

Table 53: Functional Testing 4

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	Create listing with valid data	Title, description, price, category, images provided	Listing created successfully and user is redirected to dashboard	Listing created and visible on dashboard	Pass
2	Create listing with duplicate title	Title already exists (case-insensitive)	System rejects submission with error	Duplicate title rejected correctly	Pass
3	Create listing with duplicate images	Same image uploaded twice	System rejects submission with error	Duplicate image rejected	Pass
4	Create listing with alphabetic price	Price entered as "five thousand"	System rejects submission with error	Alphabetic price rejected	Pass
5	Create listing with valid images	1–7 images, JPG/PNG format, each under 10MB	Images uploaded successfully	Images uploaded and displayed correctly	Pass

6.3.5 Functional Testing 5: Admin Approval System

Testing Objective: Ensure that the administrator can correctly approve, reject, and manage pending listings

Table 54: Functional Testing 5

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	View pending listings	Admin logged in	Display all listings with status set to "Pending"	Pending listings displayed correctly	Pass
2	Approve listing manually	Admin clicks Approve button	Listing status changes to "Active" and notification is sent to seller	Status changed to active successfully	Pass
3	Reject listing manually	Admin clicks Reject and provides reason	Listing status changes to "Rejected" and rejection reason is saved	Listing rejected with reason saved	Pass
4	Auto-approval for listings	Price within ±20% of predicted price	Listing is automatically approved without admin intervention	Auto-approved successfully	Pass
5	Manual review for listings	Price outside ±20% of predicted price	Listing is sent to admin queue for manual review	Listing sent to pending review correctly	Pass

6.3.6 Functional Testing 6: 3D AR (Augmented Reality) Features

Testing Objective: Ensure that 3D AR model upload, preview, customization, and mobile viewing work correctly for furniture listings.

Table 55: Functional Testing 6

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	Upload room photo for AR analysis	Valid JPG/PNG image, furniture listing	Room analyzed for dimensions, style, lighting; AI insights displayed	Room analysis working correctly	Pass
2	AR canvas preview with furniture	Room photo uploaded	Canvas viewer opens; furniture rendered as 3D rectangles with proper perspective	Canvas preview working correctly	Pass
3	AR customization (color/material)	Change furniture material	Furniture color/material updates in real-time on canvas	Customization working correctly	Pass
4	Drag and drop furniture placement	Drag furniture objects on canvas	Furniture repositions with shadows and perspective maintained	Drag-and-drop working correctly	Pass
5	Multiple view angles	Switch between front/3D/top views	Canvas updates to display different perspective angles	View switching working correctly	Pass

6.4 Business Rules Testing

Testing Objective: Verify that business logic rules are correctly applied for listings, price predictions, and anomalies.

Table 56: Business Rules Testing Matrix

Condition / Rule	R1	R2	R3	R4	R5
Listing Title Same?	Yes	No	No	No	No
Price unusually high?	Yes	Yes	No	No	No
Very old listing?	No	No	No	Yes	No
Invalid Input / Data Error?	No	No	No	Yes	No
Action	A1 (Flag duplicate)	A2 (Mark unusually high price)	A3 (Flag outdated listing)	A4 (Flag invalid data entry)	A5 (No anomaly)

6.5 Integration Testing

Integration testing ensures that different modules work correctly when connected together.

6.5.1 Integration Testing 1: User Registration - Email Verification - Login

Testing Objective: Verify that the user registration, authentication, and initial dashboard flow are correctly integrated.

Table 57: Integration Testing 1

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	User registers and account is created	Valid username, email, and password	User account is created in the database and a success message is displayed	Account created successfully	Pass
2	User registers with duplicate username	Username already exists	System displays error message	Duplicate username rejected	Pass
3	Newly registered user logs in	Valid newly registered credentials	User logs in successfully and is redirected to the user dashboard	Login working correctly	Pass
4	Newly registered user views empty dashboard	New user with no listings	Dashboard displays "No listings yet" message	Empty dashboard state working correctly	Pass

6.5.2 Integration Testing 2: Listing Upload - AI Price Prediction - Dashboard Update

Testing Objective: Verify the end-to-end flow from uploading a listing to predicting price and updating dashboards.

Table 58: Integration Testing 2

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	Full listing pipeline	User uploads a new listing	Data is saved successfully, price is predicted, and dashboards are updated	End-to-end process worked correctly	Pass
2	Auto-approval triggered	Listing price within approved range	Price predicted → listing auto-approved → listing becomes active in feed	Auto-approval working correctly	Pass
3	Manual review triggered	Listing price outside approved range	Price predicted → listing sent to admin → status set to pending	Manual review triggered successfully	Pass
4	User dashboard updates	User creates a new listing	Listing count increases on user dashboard	User dashboard updated correctly	Pass
5	Admin dashboard updates	New listing added to system	Total listings count increases in admin analytics	Admin dashboard updated correctly	Pass

6.5.3 Integration Testing 3: Login - Dashboard - Listings

Table 59: Integration Testing 3

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	User logs in and dashboard loads	Valid user credentials	Dashboard loads successfully with listings feed	Dashboard loaded correctly	Pass
2	User clicks Post Ad	Logged-in user	User is able to create and upload a listing	Listing upload works correctly	Pass
3	Buyer-behavior browsing	User behaving as buyer only	User can browse and scroll listings without posting	Browsing works correctly	Pass
4	User views listings	Any logged-in user	User can view all available listings	Listings displayed correctly	Pass

6.5.4 Integration Testing 4: Category Selection - Form Fields - Price Prediction

Testing Objective: Verify that dynamic form fields update correctly based on category selection.

Table 60: Integration Testing 4

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	User selects Mobile category	Category set to Mobile	Brand, RAM, storage fields are displayed	Mobile-specific fields displayed correctly	Pass
2	User enters mobile specifications	Brand: Apple, RAM: 8GB, Storage: 256GB	Predicted price is displayed along with confidence range	Price prediction working correctly	Pass
3	User selects Laptop category	Category set to Laptop	Processor, GPU, RAM, storage fields displayed	Laptop-specific fields displayed correctly	Pass
4	User changes category selection	Switch from Mobile to Laptop	Mobile fields are hidden and laptop fields are displayed	Fields updated correctly	Pass
5	User completes specifications	All required fields filled	Green auto-approval price range is displayed	Auto-approval range displayed correctly	Pass

6.5.5 Integration Testing 5: Favorites System - Dashboard - Listing Updates

Testing Objective: Verify that the favorites system is correctly integrated.

Table 61: Integration Testing 5

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	User adds a listing to favorites	User clicks favorite icon	Favorite is saved to database	Favorite saved successfully	Pass
2	User adds favorite and dashboard updates	Listing added to favorites	Favorite count increases on user dashboard	Dashboard updated correctly	Pass
3	User views favorites	User navigates to favorites page	All favorited listings are displayed	Favorites displayed correctly	Pass
4	User removes a favorite	User clicks unfavorite icon	Favorite is removed, icon changes to unfilled state	Favorite removed successfully	Pass
5	Listing deletion updates favorites	Seller deletes a listing	Listing is automatically removed from all users' favorites	Favorites cleanup working correctly	Pass

6.5.6 Integration Testing 6: 3D AR Model Upload → Storage → Preview

Testing Objective: Verify the end-to-end 3D AR workflow.

Table 62: Integration Testing 6

No.	Test Case	Attribute and Value	Expected Result	Actual Result	Result
1	Upload 3D model and store on server	Upload GLB file for furniture	Model saved to ar_previews directory	Model stored successfully	Pass
2	AR badge displayed	Furniture listing with 3D model	AR View badge visible on listing card	Badge displayed correctly	Pass
3	User clicks AR preview	Click View in AR button	AR viewer opens; 3D model renders	AR viewer working correctly	Pass
4	User customizes AR model	Change color/material and save	Customizations saved and reflected in all AR views	Customizations saved successfully	Pass

7 Conclusion and Future Work

7.1 Conclusion

EzSell successfully addresses the core challenges of the informal Pakistani second-hand economy—specifically the "trust deficit" and "pricing anxiety"—by leveraging cutting-edge Artificial Intelligence. The integration of high-performance ML ensembles for price prediction, CLIP-based computer vision for fraud prevention, and Tripo AI for immersive AR visualization has transformed a traditionally fragmented process into a secure, structured, and data-driven digital marketplace. The project demonstrates that cognitive services can bridge the gap between buyer expectations and seller reliability, ultimately creating a more transparent and efficient ecosystem for used goods.

7.2 Future Work

- In-App Escrow & Payments: Implementing a secure escrow mechanism and integrating local payment gateways like JazzCash and EasyPaisa to facilitate seamless, trust-based transactions.
- Monetization & Business Model Development: Designing a robust business model that introduces tiered services. This includes a "Freemium" approach where basic listings are free, while premium features—such as higher-visibility "Featured Ad" placement, AI-enhanced image editing, and advanced seller analytics—are available for a subscription or per-use fee.
- Dynamic Pricing Models: Introducing diverse pricing strategies tailored to different socio-economic segments, including subscription-based models for high-volume dealers and pay-as-you-go options for casual sellers to maximize platform accessibility.
- Mobile Ecosystem Expansion: Developing native iOS and Android applications using React Native or Flutter to provide a more responsive and accessible mobile experience.
- Categorical Scaling: Expanding the platform’s intelligence to complex high-value categories, including Vehicles and Real Estate, requiring specialized ML models for valuation.

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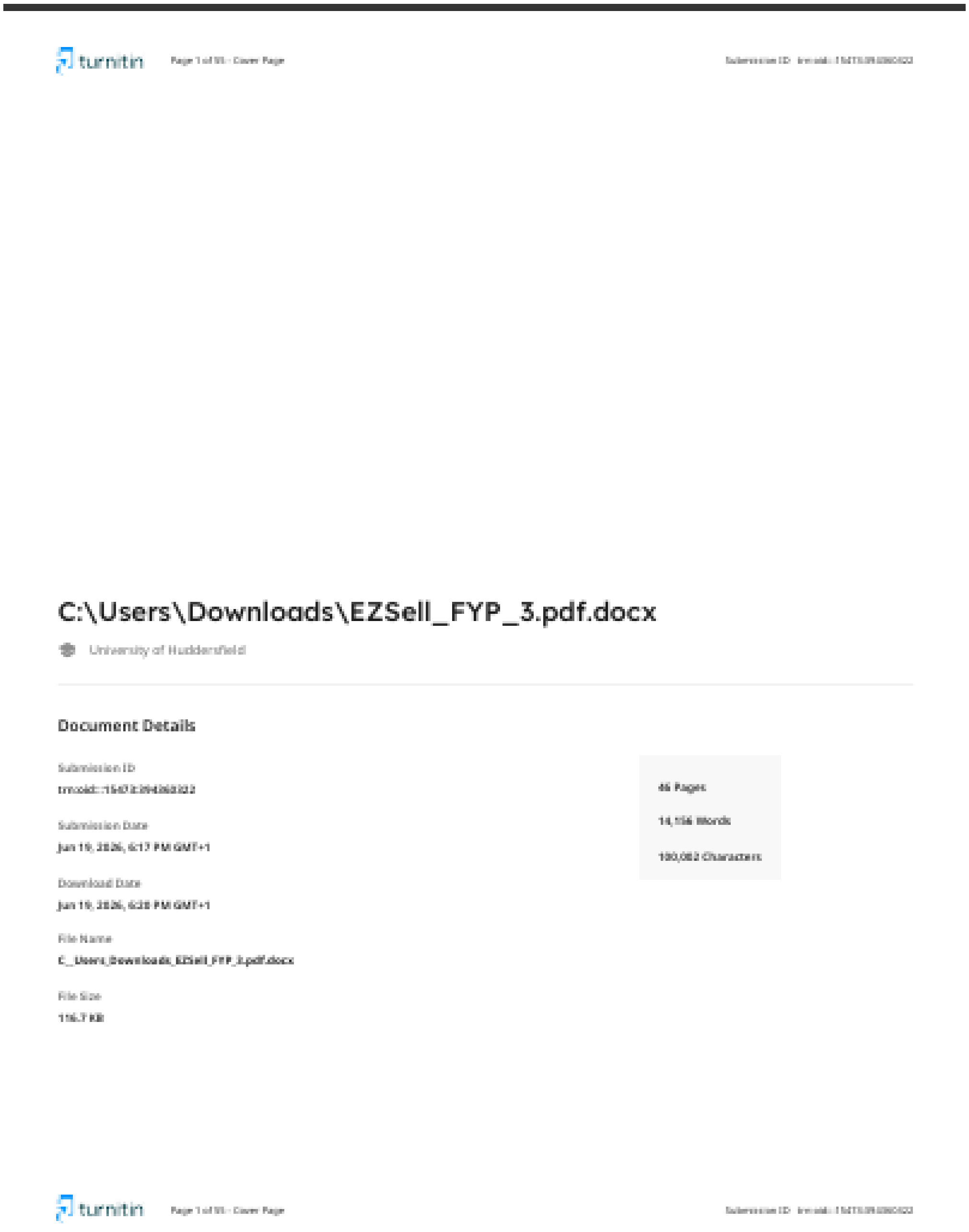

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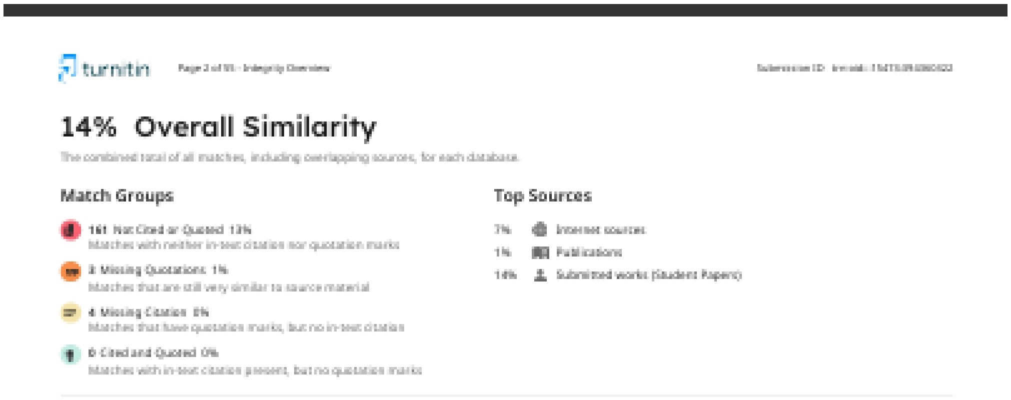

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